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Mori et al.

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(54) **PROCESSING APPARATUS AND
PROCESSING METHOD**

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(73) Assignee: **Tokyo Electron Limited**, Tokyo (JP)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 894 days.

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Jul. 18, 2019 (JP) 2019-133106

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B23K 26/53 (2014.01)

B23K 26/06 (2014.01)

(52) **U.S. Cl.**

CPC **B23K 26/53** (2015.10); **B23K 26/06** (2013.01)

(58) **Field of Classification Search**

None

See application file for complete search history.

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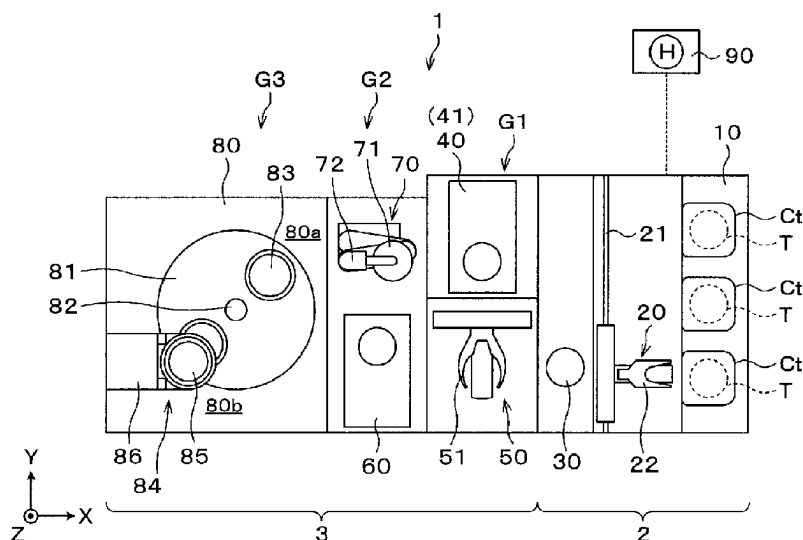
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(74) *Attorney, Agent, or Firm* — Shih IP Law Group, PLLC.

(57) **ABSTRACT**

A processing apparatus configured to process a processing target object includes a holder configured to hold the processing target object; a modifying device configured to radiate laser light to an inside of the processing target object to form multiple internal modification layers; and a controller. The controller controls an operation of forming the internal modification layers such that the internal modification layers are formed in a spiral shape from a diametrically outer side of the processing target object toward a diametrically inner side thereof, and such that an eccentric amount between the holder and the processing target object held by the holder in the forming of the internal modification layers toward the diametrically inner side of the processing target object becomes smaller than the eccentric amount in forming the internal modification layers toward the diametrically outer side of the processing target object.

11 Claims, 22 Drawing Sheets



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FIG. 1

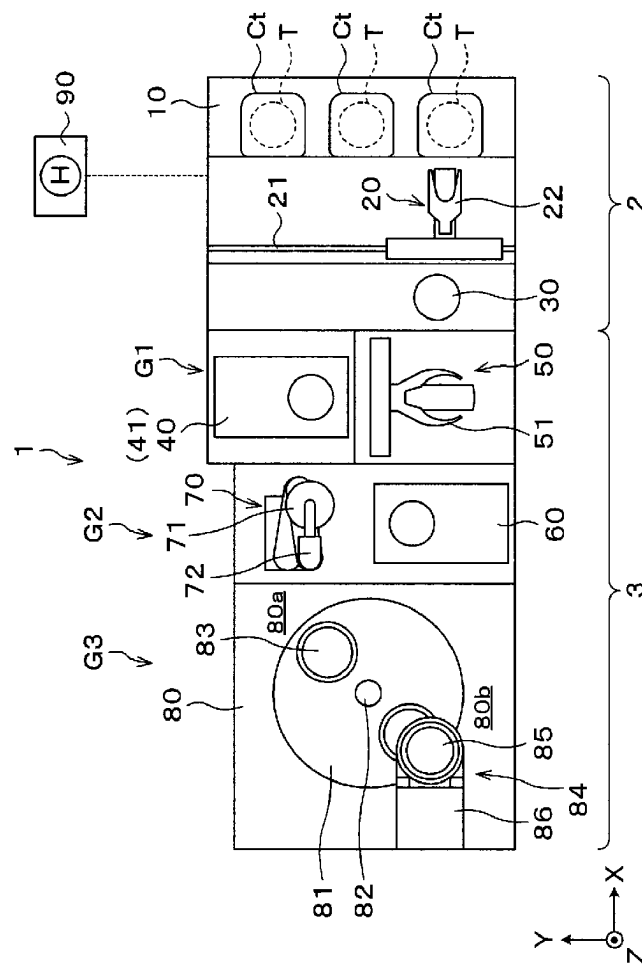


FIG. 2

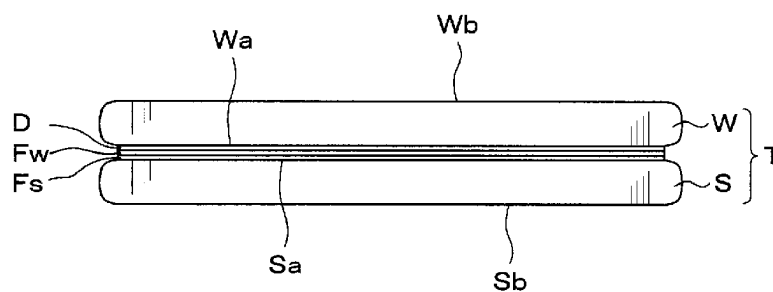


FIG. 3

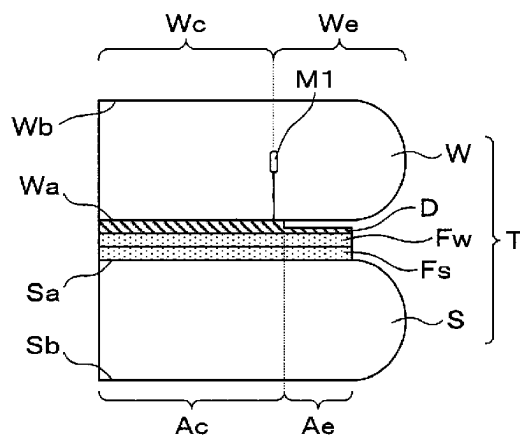


FIG. 4

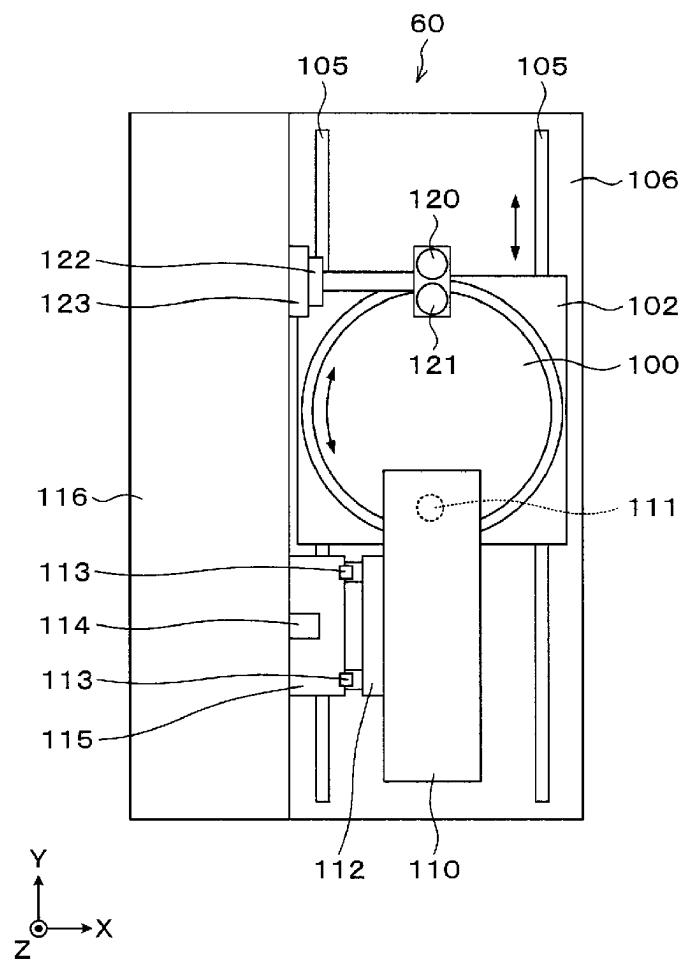


FIG. 5

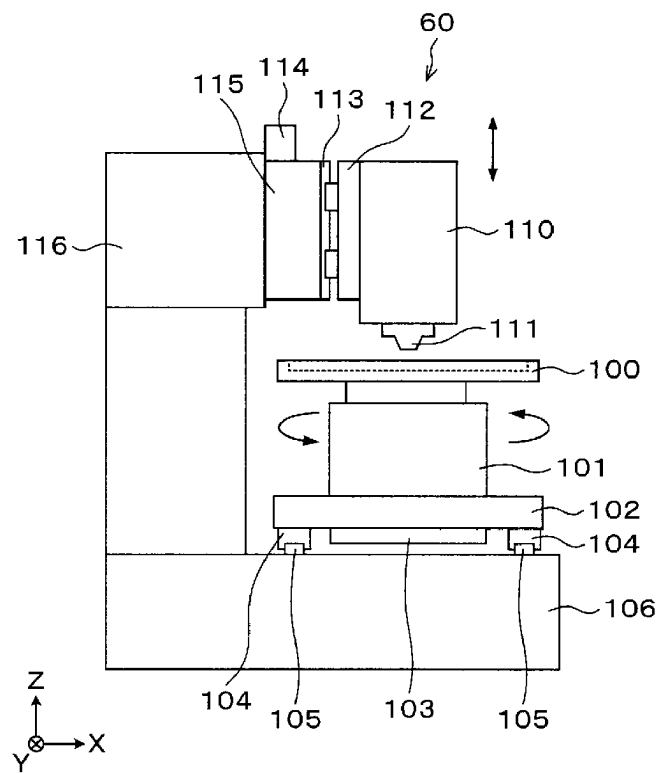
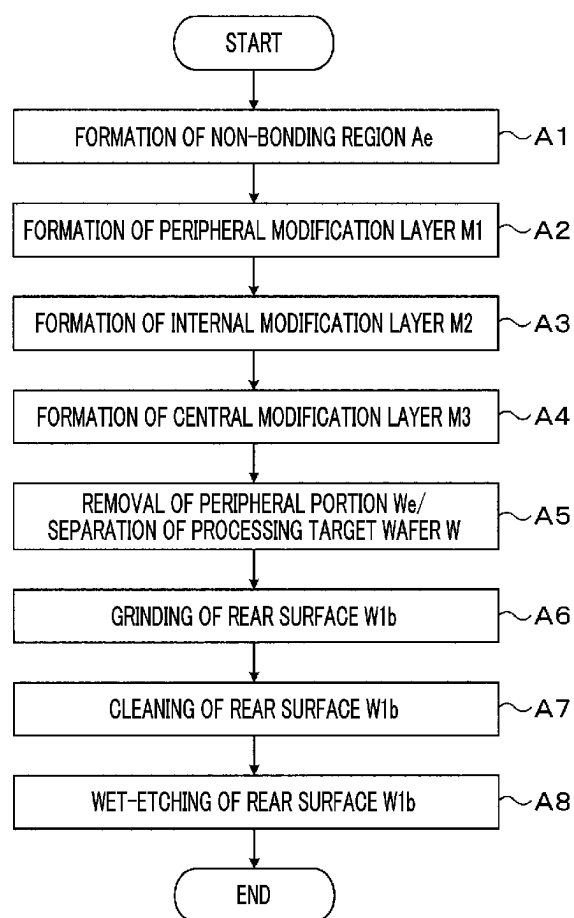


FIG. 6

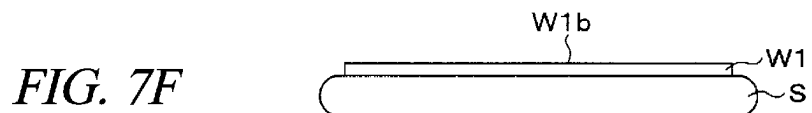
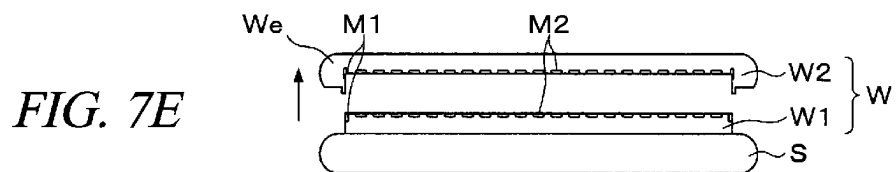
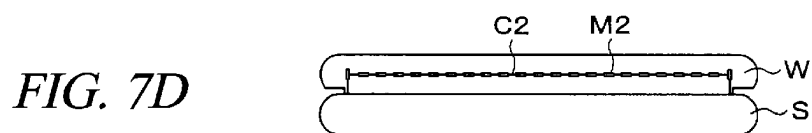
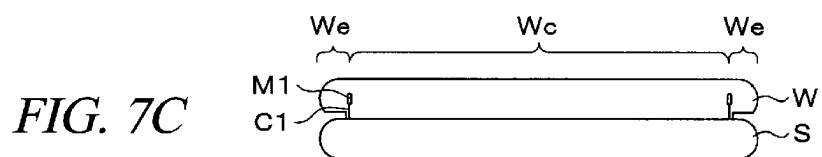
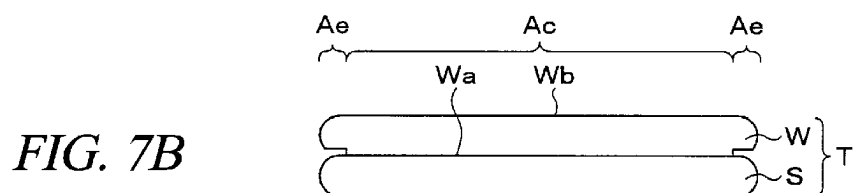
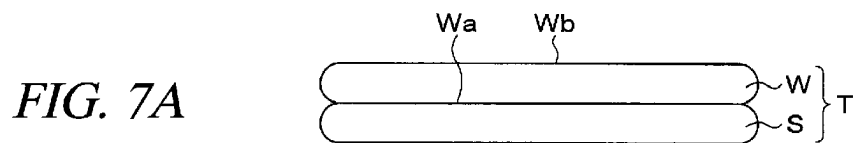


FIG. 8

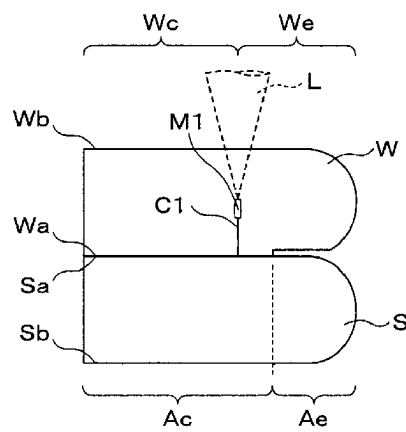


FIG. 9

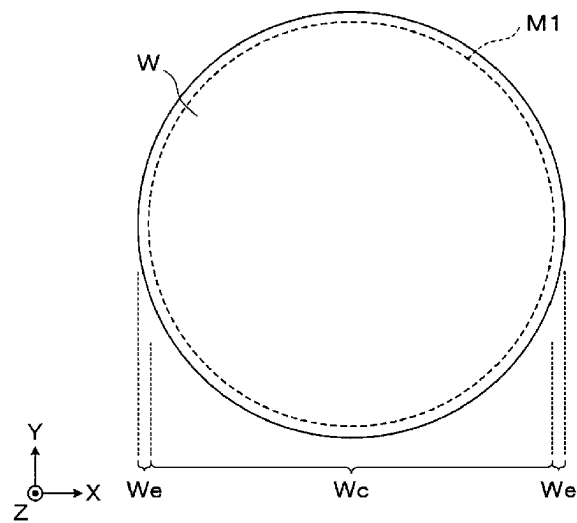


FIG. 10

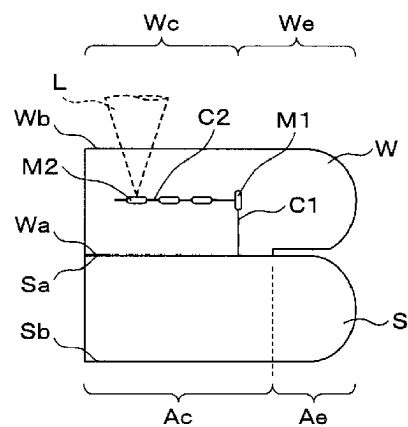


FIG. 11

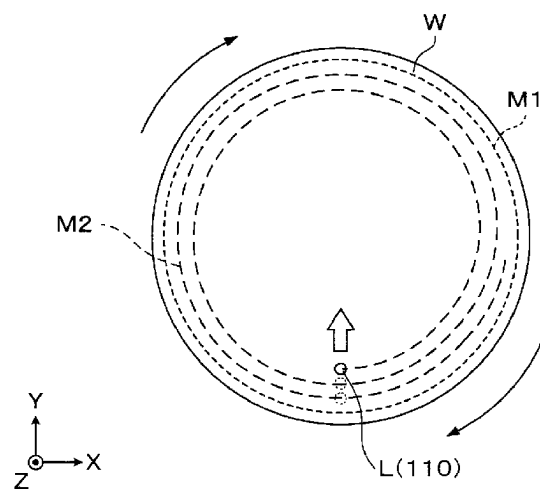


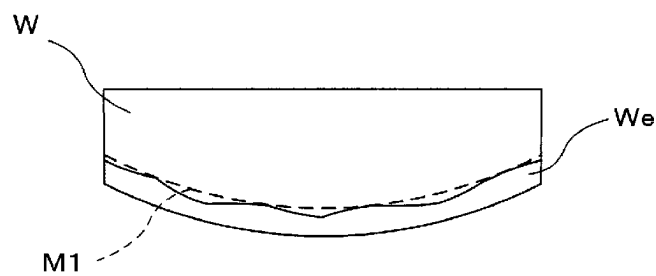
FIG. 12

FIG. 13A

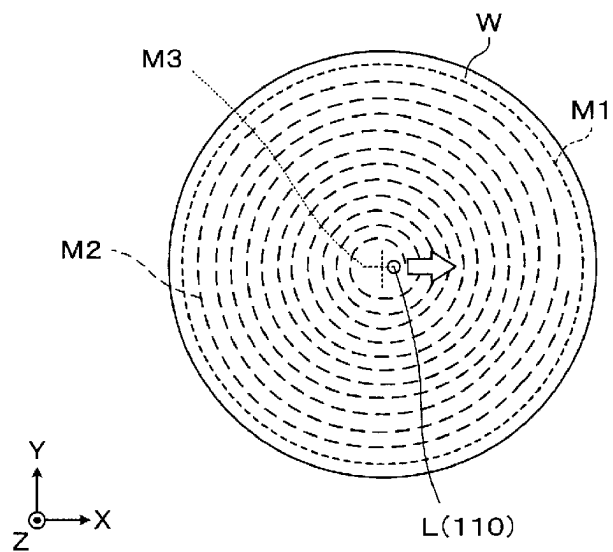


FIG. 13B

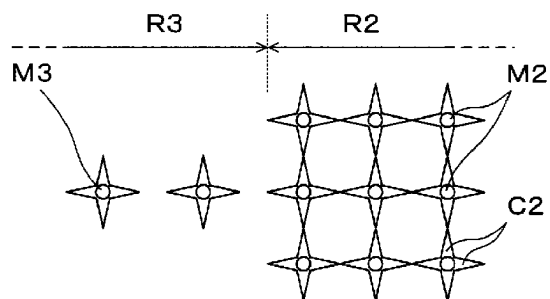


FIG. 14A

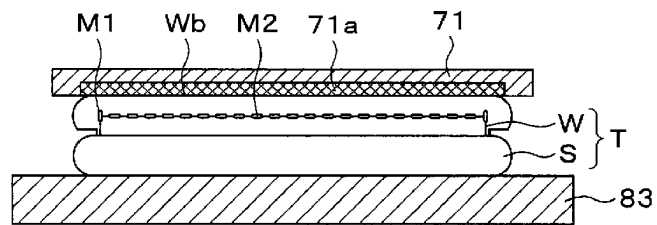


FIG. 14B

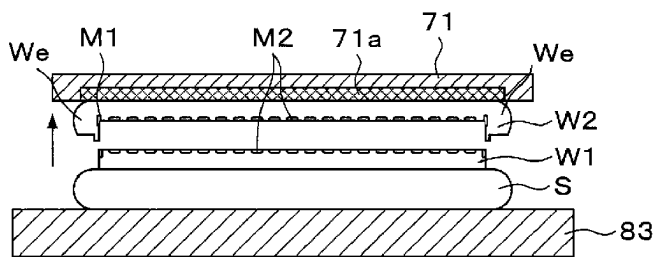


FIG. 15A

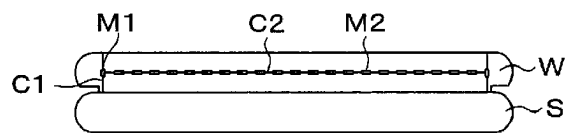


FIG. 15B

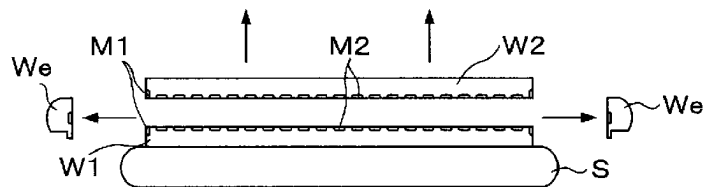


FIG. 16

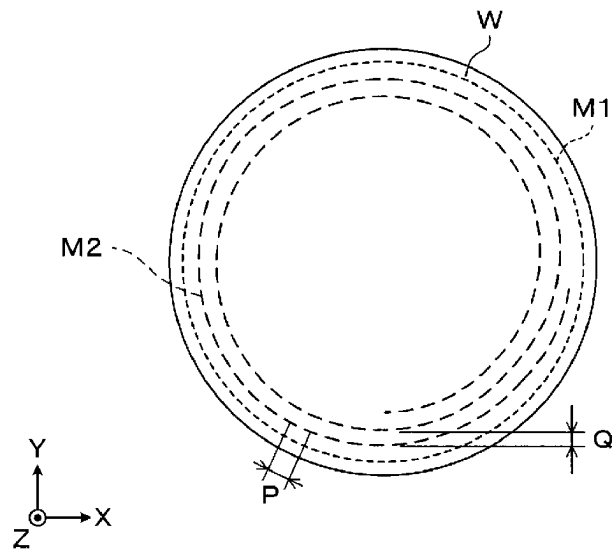


FIG. 17

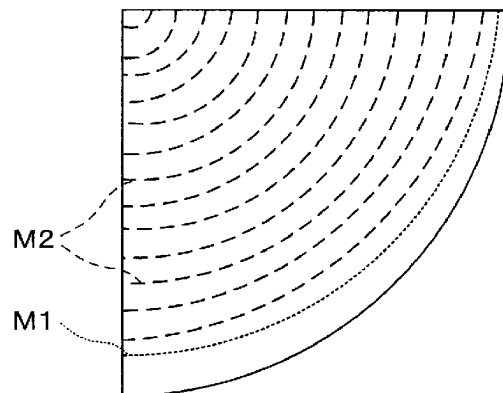


FIG. 18

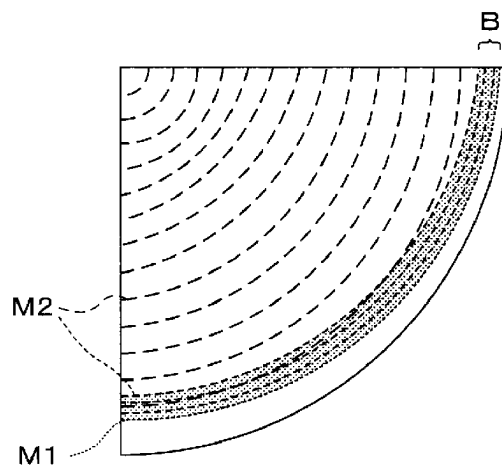


FIG. 19

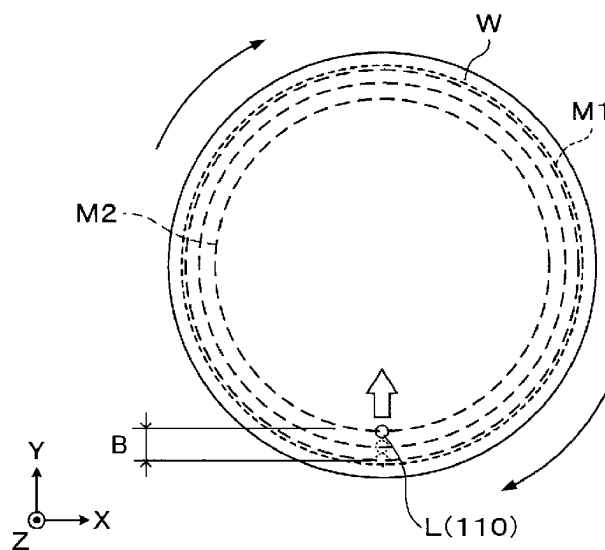


FIG. 20

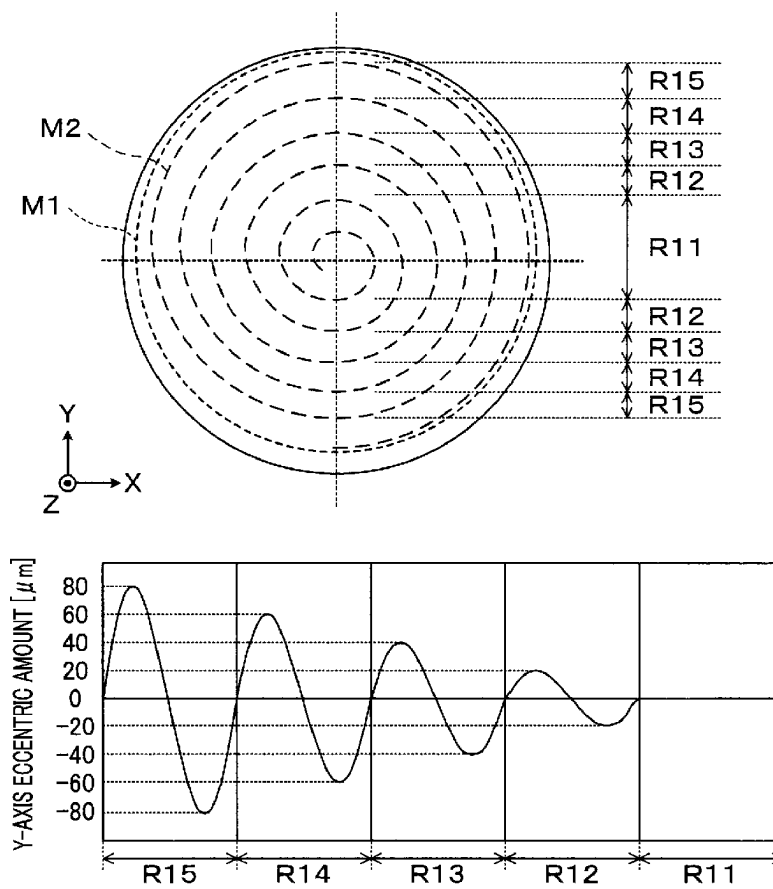


FIG. 21

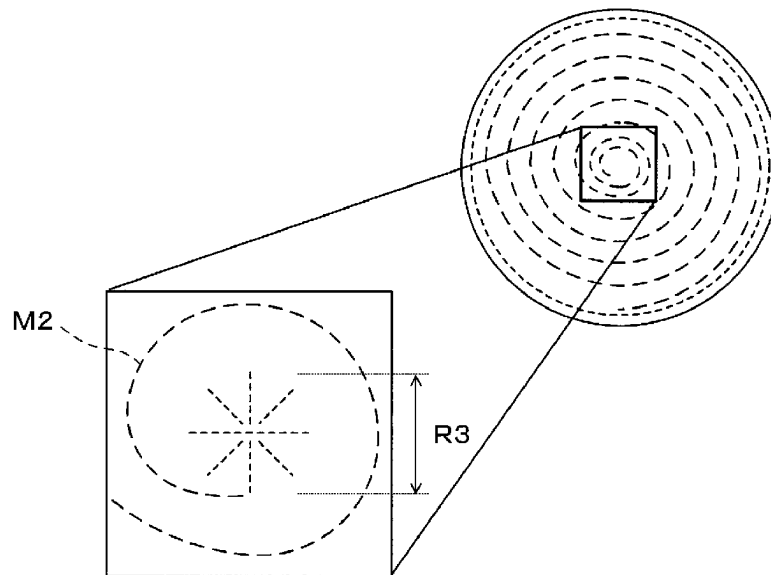


FIG. 22

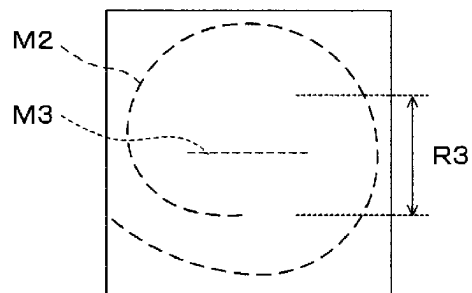


FIG. 23A

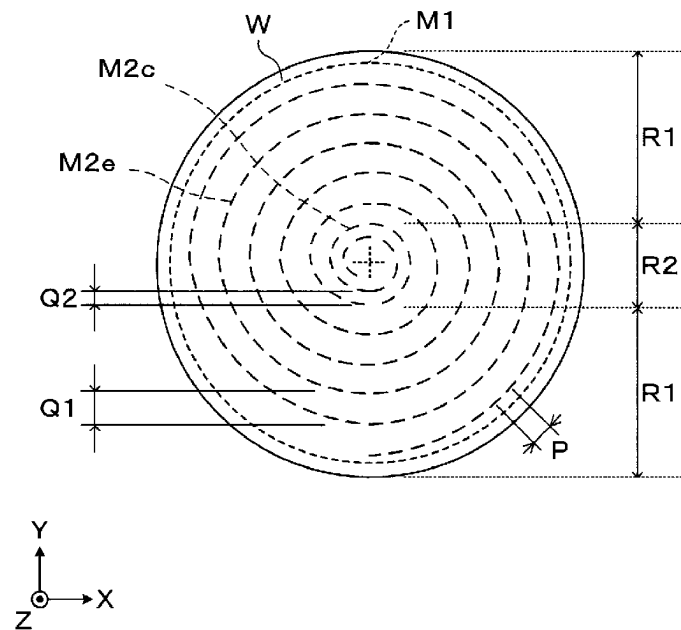


FIG. 23B

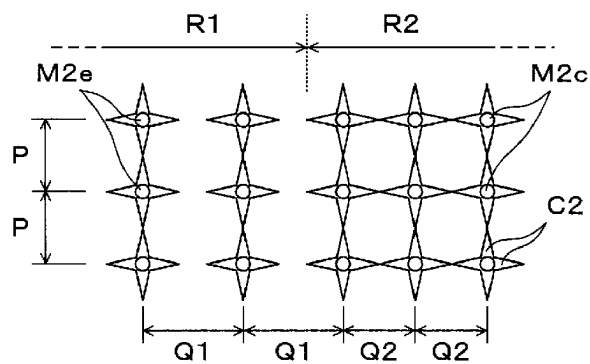


FIG. 24A

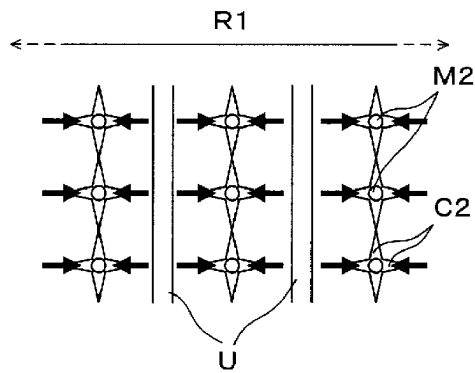


FIG. 24B

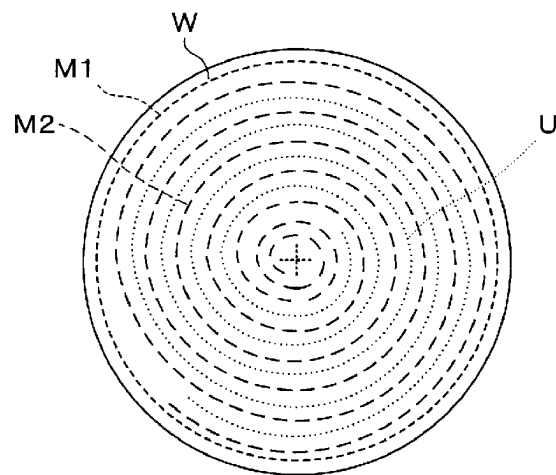


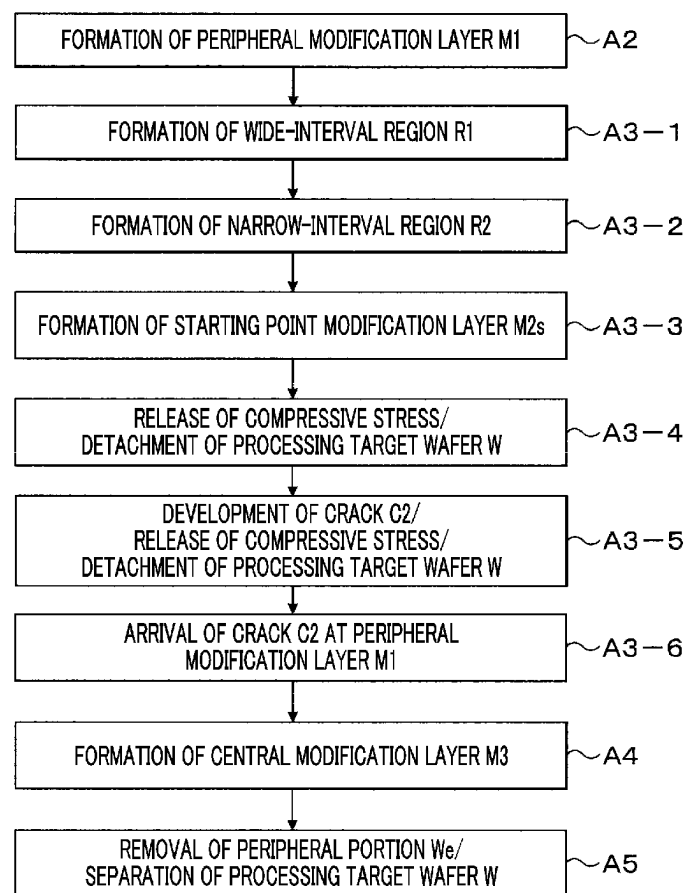
FIG. 25

FIG. 26A

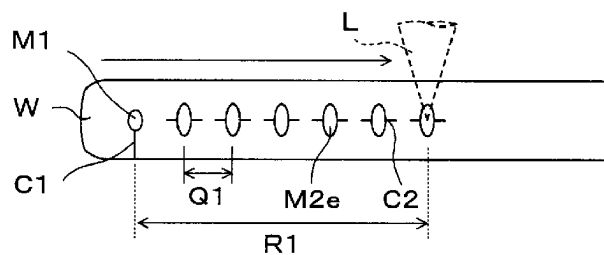


FIG. 26B

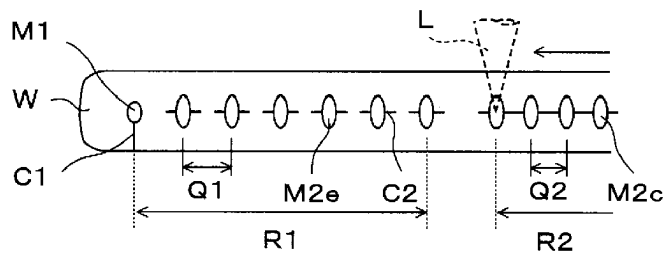


FIG. 26C

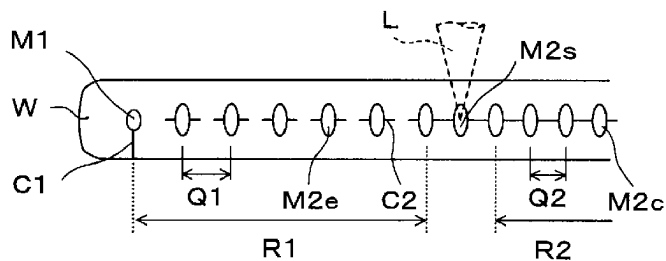


FIG. 26D

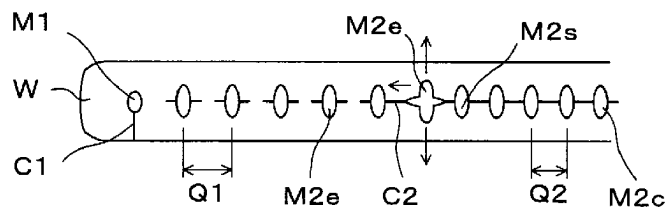


FIG. 26E

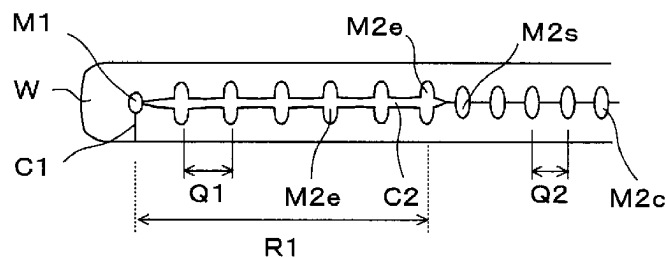


FIG. 27B

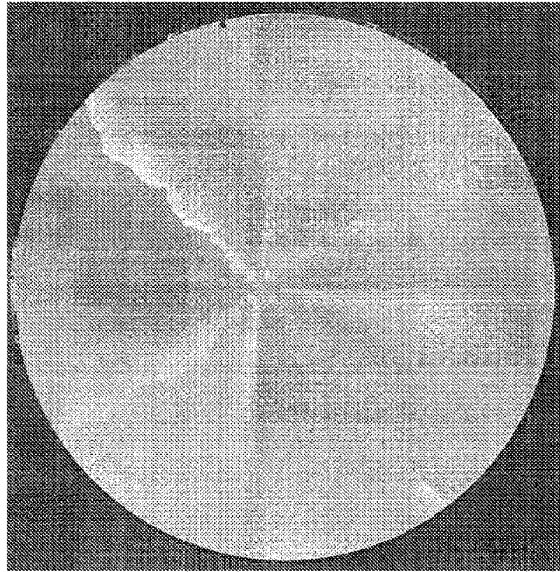


FIG. 27A

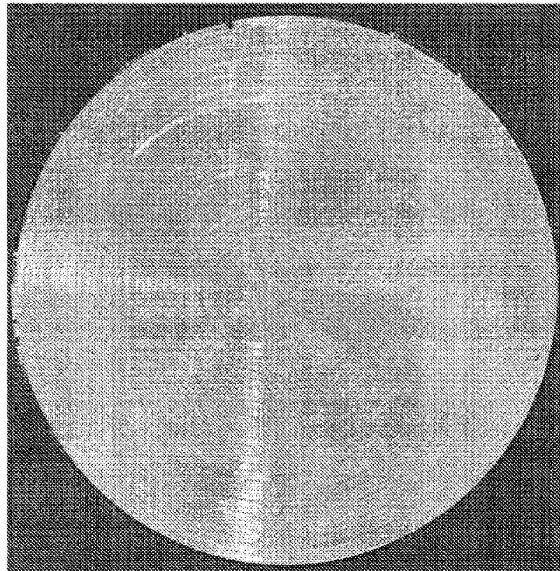


FIG. 28A

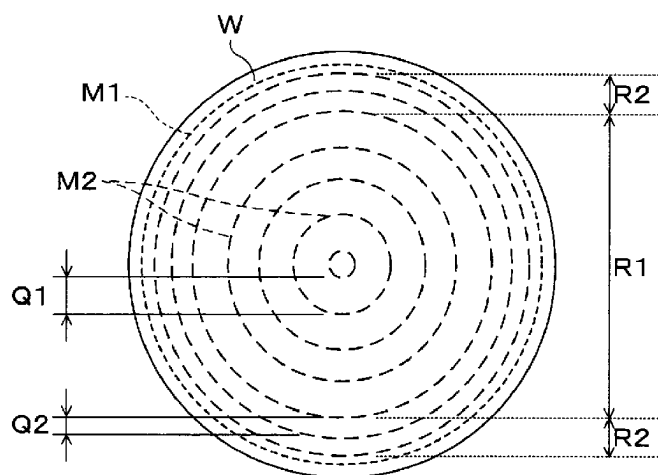


FIG. 28B

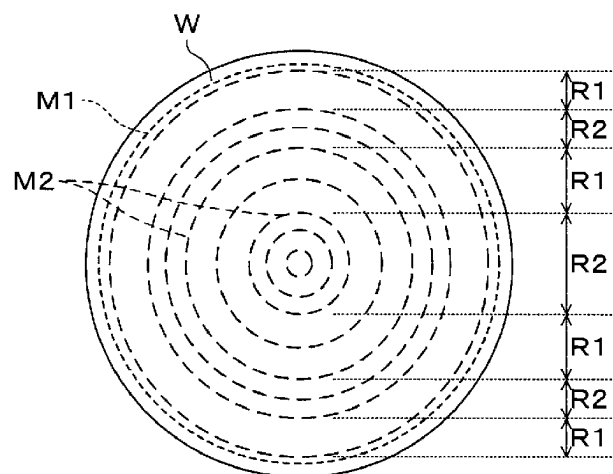


FIG. 29A

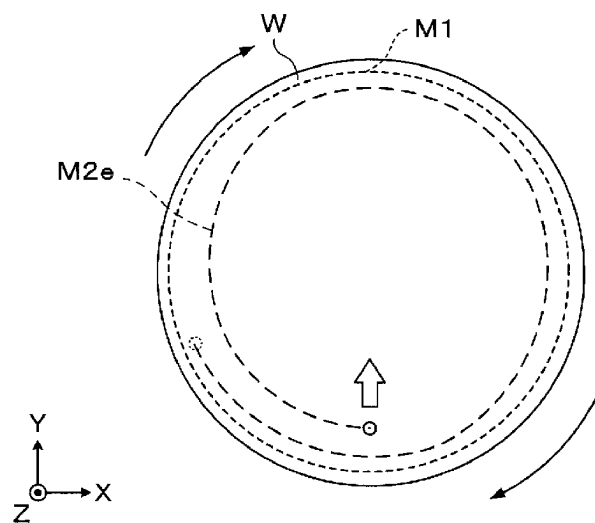
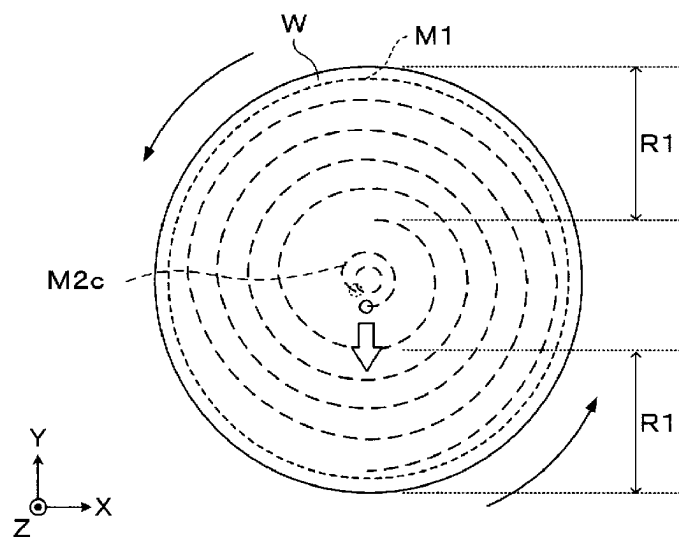


FIG. 29B



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**PROCESSING APPARATUS AND
PROCESSING METHOD****CROSS-REFERENCE TO RELATED
APPLICATIONS**

This Application is a U.S. national phase application under 35 U.S.C. § 371 of PCT Application No. PCT/JP2020/026885 filed on Jul. 9, 2020, which claims the benefit of Japanese Patent Application No. 2019-133106 filed on Jul. 18, 2019, the entire disclosures of which are incorporated herein by reference.

TECHNICAL FIELD

The various aspects and embodiments described herein pertain generally to a processing apparatus and a processing method.

BACKGROUND

Patent Document 1 discloses a method in which an internal modification layer is formed in a single crystalline substrate, and the substrate is cut using the internal modification layer as a starting point. According to Patent Document 1, the internal modification layer is formed by changing a single crystalline structure of the substrate into a polycrystalline structure while radiating laser light to an inside of the substrate. In addition, in the internal modification layer, adjacent processing traces are connected.

PRIOR ART DOCUMENT

Patent Document 1: Japanese Patent Laid-open Publication No. H2013-161820

Means for Solving the Problems

In an exemplary embodiment, a processing apparatus configured to process a processing target object includes a holder configured to hold the processing target object; a modifying device configured to radiate laser light to an inside of the processing target object to form multiple internal modification layers along a plane direction of the processing target object; and a controller configured to control an operation of forming the internal modification layers in the processing target object. The controller controls the operation of forming the internal modification layers such that the internal modification layers are formed in a spiral shape from a diametrically outer side of the processing target object toward a diametrically inner side thereof, and such that an eccentric amount between the holder and the processing target object held by the holder in the forming of the internal modification layers toward the diametrically inner side of the processing target object becomes smaller than the eccentric amount in forming the internal modification layers toward the diametrically outer side of the processing target object.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a plan view schematically illustrating a configuration example of a wafer processing system.

FIG. 2 is a side view schematically illustrating an example structure of a combined wafer.

FIG. 3 is a side view schematically illustrating an example structure of a part of the combined wafer.

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FIG. 4 is a plan view schematically illustrating a configuration example of a modifying apparatus.

FIG. 5 is a side view schematically illustrating the configuration example of the modifying apparatus.

FIG. 6 is a flowchart illustrating an example of main processes of a wafer processing.

FIG. 7A to FIG. 7F are explanatory diagrams illustrating the example of the main processes of the wafer processing.

FIG. 8 is an explanatory diagram illustrating a state in which a peripheral modification layer is being formed in a processing target wafer.

FIG. 9 is an explanatory diagram illustrating a state in which the peripheral modification layer is being formed in the processing target wafer.

FIG. 10 is an explanatory diagram illustrating a state in which an internal modification layer is being formed in the processing target wafer.

FIG. 11 is an explanatory diagram illustrating a state in which the internal modification layer is being formed in the processing target wafer.

FIG. 12 is an explanatory diagram illustrating a state in which a periphery of the processing target wafer is being removed.

FIG. 13A and FIG. 13B are explanatory diagrams illustrating a state in which a central modification layer is being formed in the processing target wafer.

FIG. 14A and FIG. 14B are explanatory diagrams illustrating a state in which the processing target wafer is being separated.

FIG. 15A and FIG. 15B are explanatory diagrams illustrating another method of separating the processing target wafer.

FIG. 16 is an explanatory diagram illustrating the formed internal modification layer.

FIG. 17 is an explanatory diagram illustrating a first eccentricity correction method.

FIG. 18 is an explanatory diagram illustrating a second eccentricity correction method.

FIG. 19 is an explanatory diagram illustrating a method of forming the internal modification layer in the second eccentricity correction method.

FIG. 20 is an explanatory diagram illustrating a third eccentricity correction method.

FIG. 21 is an explanatory diagram illustrating a state in which the central modification layer is being formed in the processing target wafer.

FIG. 22 is an explanatory diagram illustrating another example of forming the central modification layer.

FIG. 23A and FIG. 23B are explanatory diagrams illustrating a method of forming an internal modification layer according to a second exemplary embodiment.

FIG. 24A and FIG. 24B are explanatory diagrams illustrating the internal modification layer formed in the second exemplary embodiment.

FIG. 25 is a flowchart illustrating example processes of a wafer processing according to the second exemplary embodiment.

FIG. 26A to FIG. 26E are explanatory diagrams illustrating the example processes of the wafer processing according to the second exemplary embodiment.

FIG. 27A and FIG. 27B are explanatory diagrams illustrating improvement of surface roughness of the processing target wafer.

FIG. 28A and FIG. 28B are explanatory diagrams illustrating other examples of the internal modification layer formed according to the second exemplary embodiment.

FIG. 29A and FIG. 29B are explanatory diagrams illustrating other examples of the internal modification layer formed according to the second exemplary embodiment.

DETAILED DESCRIPTION

In a manufacturing process for a semiconductor device, a semiconductor wafer (hereinafter, simply referred to as a wafer) such as a circular substrate having a plurality of devices such as electronic circuits formed on a surface thereof is thinned by radiating laser light to an inside of the wafer to form a modification layer and separating the wafer by using the modification layer as a starting point, as described in Patent Document 1, for example.

For such wafer separation, after the modification layer is formed within the wafer, a tensile force in a detaching direction is applied while holding a front side and a rear side of the wafer. Accordingly, the wafer is separated and thinned using the formed modification layer and cracks propagating from the modification layer as a boundary. In the following description, among separated wafers, the wafer on the front side where the devices are formed will be sometimes referred to as a “first separation wafer,” and the wafer on the rear side, as a “second separation wafer”.

Further, in the separation of the wafer, an edge trimming processing is performed to suppress a peripheral portion of the wafer from having a sharp pointed shape (a so-called knife edge shape) by the separation. In the edge trimming processing, a modification layer is formed by radiating laser light to an inside of the wafer along a peripheral portion of the wafer as a removing target, and the peripheral portion is detached starting from this modification layer.

However, when performing the separation of the wafer and the removal of the peripheral portion thereof as described above, if the modification layer is formed eccentrically with respect to the wafer, the separation of the wafer and the removal of the peripheral portion thereof may not be performed properly. To be specific, if the modification layer for the detachment is eccentric toward a peripheral side of the wafer, the quality of the removal of the peripheral portion may not be maintained, whereas if the modification layer is eccentric toward a center side of the wafer, the separation may not be performed properly. Patent Document 1 does not mention anything about this eccentricity of the modification layer. Thus, there is a room for improvement.

The present disclosure provides a technique enabling to perform a periphery removing processing and a separation processing for a processing target object appropriately. Hereinafter, a wafer processing system equipped with a processing apparatus according to an exemplary embodiment and a wafer processing method as a processing method will be described with reference to the accompanying drawings. In the present specification and the drawings, parts having substantially the same functions and configurations will be assigned same reference numerals, and redundant description thereof will be omitted.

First, a configuration of the wafer processing system will be discussed. FIG. 1 is a plan view schematically illustrating a configuration of a wafer processing system 1.

The wafer processing system 1 is configured to perform a processing on a combined wafer T in which a processing target wafer W and a support wafer S are bonded to each other as illustrated in FIG. 2. In the wafer processing system 1, the processing target wafer W is separated and thinned. Hereinafter, in the processing target wafer W, a surface bonded to the support wafer S will be referred to as a front surface Wa, and a surface opposite to the front surface Wa

will be referred to as a rear surface Wb. Likewise, in the support wafer S, a surface bonded to the processing target wafer W will be referred to as a front surface Sa, and a surface opposite to the front surface Sa will be referred to as a rear surface Sb. Further, in the present exemplary embodiment, the processing target wafer W corresponds to a processing target object of the present disclosure.

The processing target wafer W is a semiconductor wafer such as, but not limited to, a silicon wafer having a circular plate shape, and it has, on the front surface Wa thereof, a device layer D including a plurality of devices such as electronic circuits. Further, an oxide film Fw, for example, a SiO₂ film (TEOS film) is further formed on the device layer D. In the present exemplary embodiment, the processing target wafer W constitutes the aforementioned wafer as the target of separation.

The support wafer S is a wafer that supports the processing target wafer W. An oxide film Fs, for example, a SiO₂ film (TEOS film) is formed on the front surface Sa of the support wafer S. Further, if the support wafer S has a plurality of devices formed on the front surface Sa thereof, a device layer (not shown) is formed on the front surface Sa, the same as in the processing target wafer W.

Further, in the following description, illustration of the device layer D and the oxide films Fw and Fs may be omitted in some cases for the simplicity of illustration.

Further, in addition to the above-stated thinning processing, an edge trimming processing is further performed on the processing target wafer W to suppress a peripheral portion of the processing target wafer W from having a knife edge shape by the thinning processing as described above. In the edge trimming processing, as shown in FIG. 3, a peripheral modification layer M1 is formed by radiating laser light to a boundary between a peripheral portion We as a removing target and a central portion Wc, and the peripheral portion We is removed starting from this peripheral modification layer M1. Further, the peripheral portion We to be removed by the edge trimming may range from, e.g., 1 mm to 5 mm from an edge of the processing target wafer W in a diametrical direction thereof. A method of the edge trimming processing will be described later.

Here, if the processing target wafer W and the support wafer S are bonded in the peripheral portion We of the processing target wafer W, there is a likelihood that the peripheral portion We may not be removed appropriately. For the reason, a non-bonding region Ae for appropriately performing the edge trimming is formed at an interface between the processing target wafer W and the support wafer S in a portion corresponding to the peripheral portion We as the removal target in the edge trimming. Specifically, as shown in FIG. 3, a bonding region Ac in which the processing target wafer W and the support wafer S are bonded and the non-bonding region Ae in which bonding strength between the processing target wafer W and the support wafer S is reduced are formed at the interface between the processing target wafer W and the support wafer S. Further, it is desirable that an outer end of the bonding region Ac is located slightly outer than an inner end of the peripheral portion We to be removed.

The non-bonding region Ae may be formed before the bonding, for example. Specifically, by removing a bonding interface of the processing target wafer W before being subjected to the bonding through polishing or wet etching, by modifying the bonding interface through radiation of laser light thereto, or by hydrophobizing the bonding interface through application of a hydrophobic material thereon, the bonding strength is reduced to form the non-bonding

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region Ae. Further, the “bonding interface” where the non-bonding region Ae is formed refers to a portion of the processing target wafer W forming an interface to be actually bonded to the support wafer S.

The non-bonding region Ae may be formed after the bonding, for example. Specifically, by radiating laser light to the interface in a portion corresponding to the peripheral portion We of the processing target wafer W after the bonding, the bonding strength for the front surface Sa of the support wafer S is reduced, so that the non-bonding region Ae is formed. In addition, the non-bonding region Ae may be formed at any position in the vicinity of the bonding interface between the processing target wafer W and the support wafer S as long as a bonding force between the processing target wafer W and the support wafer S in the vicinity of the bonding interface” according to the present exemplary embodiment includes the inside of the processing target wafer W, the inside of the device layer D, the inside of an oxide film Fw, and so forth.

As depicted in FIG. 1, the wafer processing system 1 includes a carry-in/out station 2 and a processing station 3 connected as one body. In the carry-in/out station 2, a cassette Ct capable of accommodating therein a multiple number of combined wafers T is carried to/from the outside, for example. The processing station 3 is equipped with various kinds of processing apparatuses configured to perform required processings on the combined wafers T.

A cassette placing table 10 is provided in the carry-in/out station 2. In the shown example, a plurality of, for example, three cassettes Ct can be arranged on the cassette placing table 10 in a row in the Y-axis direction. Further, the number of the cassettes Ct placed on the cassette placing table 10 is not limited to the example of the present exemplary embodiment but can be selected as required.

In the carry-in/out station 2, a wafer transfer device 20 is provided adjacent to the cassette placing table 10 at a negative X-axis side of the cassette placing table 10. The wafer transfer device 20 is configured to be movable on a transfer path 21 which is elongated in the Y-axis direction. Further, the wafer transfer device 20 is equipped with, for example, two transfer arms 22 each of which is configured to hold and transfer the combined wafer T. Each transfer arm 22 is configured to be movable in a horizontal direction and a vertical direction and pivotable around a horizontal axis and a vertical axis. Further, the configuration of the transfer arm 22 is not limited to the exemplary embodiment, and various other configurations may be adopted. The wafer transfer device 20 is configured to be capable of transferring the combined wafer T to/from the cassette Ct of the cassette placing table 10 and a transition device 30 to be described later.

In the carry-in/out station 2, the transition device 30 configured to deliver the combined wafer T is provided adjacent to the wafer transfer device 20 at a negative X-axis side of the wafer transfer device 20.

The processing station 3 is provided with, for example, three processing blocks G1 to G3. The first processing block G1, the second processing block G2 and the third processing block G3 are arranged side by side in this sequence from a positive X-axis side (from the carry-in/out station 2 side) toward a negative X-axis side.

The first processing block G1 is equipped with an etching apparatus 40, a cleaning apparatus 41, and a wafer transfer device 50. The etching apparatus 40 and the cleaning apparatus 41 are stacked on top of each other. Further, the

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number and the layout of the etching apparatus 40 and the cleaning apparatus 41 are not limited to the shown example. By way of example, the etching apparatus 40 and the cleaning apparatus 41 may be arranged side by side in the X-axis direction. Further, a plurality of etching apparatuses 40 and a plurality of cleaning apparatuses 41 may be respectively stacked on top of each other.

The etching apparatus 40 is configured to etch a separated surface of the processing target wafer W grounded by a processing apparatus 80 to be described later. By way of example, by supplying a chemical liquid (etching liquid) onto the separated surface, this separated surface is wet-etched. For instance, HF, HNO₃, H₃PO₄, TMAH, Choline, KOH, or the like may be used as the chemical liquid.

The cleaning apparatus 41 is configured to clean the separated surface of the processing target wafer W grounded by the processing apparatus 80 to be described later. By way of example, by bringing a brush into contact with the separated surface, the separated surface is cleaned by being scrubbed. Furthermore, a pressurized cleaning liquid may be used for the cleaning of the separated surface. In addition, the cleaning apparatus 41 may be configured to clean the rear surface Sb of the support wafer S as well as the separated surface of the processing target wafer W.

The wafer transfer device 50 is disposed at, for example, a negative Y-axis side of the etching apparatus 40 and the cleaning apparatus 41. The wafer transfer device 50 has, for example, two transfer arms 51 each of which is configured to hold and transfer the combined wafer T. Each transfer arm 51 is configured to be movable in a horizontal direction and a vertical direction and pivotable around a horizontal axis and a vertical axis. Further, the configuration of the transfer arm 51 is not limited to the exemplary embodiment, and various other configurations may be adopted. Additionally, the wafer transfer device 50 is configured to be capable of transferring the combined wafer T to/from the transition device 30, the etching apparatus 40, the cleaning apparatus 41 and a modifying apparatus 60 to be described later.

The second processing block G2 is equipped with the modifying apparatus 60 as a processing apparatus and a wafer transfer device 70. The number and the layout of the modifying apparatus 60 is not limited to the example of the present exemplary embodiment, and a plurality of modifying apparatuses 60 may be stacked.

The modifying apparatus 60 is configured to form the non-bonding region Ae, the peripheral modification layer M1, an internal modification layer M2, and a central modification layer M3 by radiating laser light to an inside of the processing target wafer W. A detailed configuration of the modifying apparatus 60 will be elaborated later.

The wafer transfer device 70 is disposed at, for example, a positive Y-axis side of the modifying apparatus 60. The wafer transfer device 70 is equipped with, for example, two transfer arms 71 each of which is configured to hold and transfer the combined wafer T. Each transfer arm 71 is supported at a multi-joint arm member 72 and configured to be movable in a horizontal direction and a vertical direction and pivotable around a horizontal axis and a vertical axis. Further, the configuration of the transfer arm 71 is not limited to the example of the present exemplary embodiment, and may vary as required. The wafer transfer device 70 is configured to be capable of transferring the combined wafer T to/from the cleaning apparatus 41, the modifying apparatus 60, and the processing apparatus 80 to be described later.

The third processing block G3 is equipped with the processing apparatus 80. The number and the layout of the

processing apparatus **80** is not limited to the example of the present exemplary embodiment, and a plurality of processing apparatuses **80** may be arranged as required.

The processing apparatus **80** has a rotary table **81**. The rotary table **81** is configured to be rotatable about a vertical rotation center line **82** by a rotation mechanism (not shown). Two chucks **83** each configured to attract and hold the combined wafer **T** are provided on the rotary table **81**. The chucks **83** are arranged on a circle concentric with the rotary table **81** in a uniform manner. The two chucks **83** are configured to be moved to a delivery position **80a** and a processing position **80b** as the rotary table **81** is rotated. Further, each of the two chucks **83** is configured to be rotatable around a vertical axis by a rotating mechanism (not shown).

At the delivery position **80a**, delivery of the combined wafer **T** is performed. The grinding unit **84** is disposed at the processing position **80b** to grind the processing target wafer **W**. The grinding unit **84** is equipped with a grinder **85** having a grinding whetstone (not shown) configured to be rotated in a ring shape. Further, the grinder **85** is configured to be movable in a vertical direction along a supporting column **86**. While keeping the processing target wafer **W** held by the chuck **83** in contact with the grinding whetstone, the chuck **83** and the grinding whetstone are respectively rotated.

The above-described wafer processing system **1** is equipped with a control device **90** as a controller. The control device **90** is implemented by, for example, a computer equipped with a CPU, a memory, and so forth, and includes a program storage (not shown). A program for controlling a processing of the processing target wafer **W** in the wafer processing system **1** is stored in the program storage. Further, the program storage also stores therein a program for implementing a wafer processing to be described later in the wafer processing system **1** by controlling the above-described various processing apparatuses and a driving system such as the transfer devices. Further, the programs may be recorded in a computer-readable recording medium **H**, and may be installed from this recording medium **H** to the control device **90**.

Moreover, the aforementioned various processing apparatus may be further equipped with control devices (not shown) respectively configured to control the various processing apparatuses independently.

Now, the aforementioned modifying apparatus **60** will be described. FIG. 4 and FIG. 5 are a plan view and a side view illustrating a schematic configuration of the modifying apparatus **60**, respectively.

The modifying apparatus **60** is equipped with a chuck **100** as a holder configured to hold the combined wafer **T** on a top surface thereof. The chuck **100** is configured to attract and hold the rear surface **Sb** of the support wafer **S** in the state that the processing target wafer **W** is placed at an upper side and the support wafer **S** is placed at a lower side. The chuck **100** is supported on a slider table **102** with an air bearing **101** therebetween. A rotating mechanism **103** is provided at a bottom surface side of the slider table **102**. The rotating mechanism **103** incorporates therein, for example, a motor as a driving source. The chuck **100** is configured to be rotated around a vertical axis by the rotating mechanism **103** via the air bearing **101** therebetween. The slider table **102** is configured to be moved by a moving mechanism **104**, which is provided at a bottom surface side of the slider table **102** to serve as a holder moving mechanism, along a rail **105** which is provided on a base **106** and elongated in the Y-axis

direction. Further, though not particularly limited, a driving source of the moving mechanism **104** may be, for example, a linear motor.

A laser head **110** serving as a modifying device is provided above the chuck **100**. The laser head **110** has a lens **111**. The lens **111** is a cylindrical member provided on a bottom surface of the laser head **110**, and is configured to radiate the laser light to the processing target wafer **W** held by the chuck **100**.

The laser head **110** is configured to concentrate and radiate the laser light having a wavelength featuring transmissivity for the processing target wafer **W** to a preset position within the processing target wafer **W** as high-frequency laser light in a pulse shape oscillated from a laser light oscillator (not shown). Accordingly, a portion within the processing target wafer **W** to which the laser light is concentrated is modified, so that the non-bonding region **Ae**, the peripheral modification layer **M1**, the internal modification layer **M2**, and the central modification layer **M3** are formed.

In the present exemplary embodiment, in order to avoid complication of illustration, it is assumed that the non-bonding region **Ae**, the peripheral modification layer **M1**, the internal modification layer **M2**, and the central modification layer **M3** are formed by the common laser head **110**. However, they may be formed by different laser heads. In addition, the laser heads may be used selectively depending on the type of laser light to be radiated.

The laser head **110** is supported at a supporting member **112**. The laser head **110** is configured to be moved up and down by an elevating mechanism **114** along a vertically elongated rail **113**. Further, the laser head **110** is configured to be moved in the Y-axis direction by a moving mechanism **115** as a modifying device moving mechanism. Each of the elevating mechanism **114** and the moving mechanism **115** is supported at a supporting column **116**.

Above the chuck **100**, a macro-camera **120** and a micro-camera **121** are provided at a positive Y-axis side of the laser head **110**. For example, the macro-camera **120** and the micro-camera **121** are formed as one body, and the macro-camera **120** is provided at a positive Y-axis side of the micro-camera **121**. The macro-camera **120** and the micro-camera **121** are configured to be moved up and down by an elevating mechanism **122**, and also configured to be moved in the Y-axis direction by a moving mechanism **123**.

The macro-camera **120** is configured to image an outer end portion of the processing target wafer **W** (combined wafer **T**). The macro-camera **120** is equipped with, for example, a coaxial lens, and radiates visible light, for example, red light and receives reflection light from a target object. For example, the macro-camera **120** has an image magnification of two times.

The image obtained by the macro-camera **120** is outputted to the control device **90**. The control device **90** calculates a first eccentric amount between a center of the chuck **100** and a center of the processing target wafer **W** from the image obtained by the macro camera **120**.

The micro-camera **121** is configured to image a peripheral portion of the processing target wafer **W** and image a boundary between the bonding region **Ac** and the non-bonding region **Ae**. The micro-camera **121** is equipped with, for example, a coaxial lens, and radiates infrared light (IR light) and receives reflection light from a target object. By way of example, the micro-camera **121** has an image magnification of 10 times. A field of view of the micro-camera **121** is about $\frac{1}{5}$ of a field of view of the macro-camera **120**,

and a pixel size of the micro-camera **121** is about $\frac{1}{5}$ of a pixel size of the macro-camera **120**.

The image obtained by the micro-camera **121** is outputted to the control device **90**. In the control device **90**, a second eccentric amount between the center of the chuck **100** and the center of the bonding region **Ac** is calculated from the image obtained by the micro-camera **121**. Also, the control device **90** moves the chuck **100** or the laser head **110** based on the second eccentric amount so that the center of the chuck **100** and the center of the bonding region **Ac** are coincident with each other. In the following description, this control of moving the chuck **100** or the laser head **110** will be sometimes referred to as eccentricity correction.

Now, a wafer processing performed by using the wafer processing system **1** configured as described above will be discussed. FIG. **6** is a flowchart illustrating main processes of the wafer processing. FIG. **7A** to FIG. **7F** are explanatory diagrams illustrating the main processes of the wafer processing. In the present exemplary embodiment, the combined wafer **T** is previously formed by bonding the processing target wafer **W** and the support wafer **S** in a bonding apparatus (not shown) at the outside of the wafer processing system **1**. Further, although the combined wafer **T** carried into the wafer processing system **1** is already provided with the aforementioned non-bonding region **Ae** formed thereat, the following description will be provided for an example where the non-bonding region **Ae** is formed in the modifying apparatus **60**.

First, the cassette **Ct** accommodating therein the multiple number of combined wafers **T** shown in FIG. **7A** is placed on the cassette placing table **10** of the carry-in/out station **2**.

Then, the combined wafer **T** is taken out of the cassette **Ct** by the wafer transfer device **20**, and transferred into the transition device **30**. Subsequently, the combined wafer **T** is taken out of the transition device **30** by the wafer transfer device **50**, and transferred into the modifying apparatus **60**. In the modifying apparatus **60**, the non-bonding region **Ae** is first formed, as shown in FIG. **7B** (process **A1** of FIG. **6**). Subsequently, a peripheral modification layer **M1** is formed inside the processing target wafer **W**, as illustrated in FIG. **7C** (process **A2** of FIG. **6**), an internal modification layer **M2** is formed, as shown in FIG. **7D** (process **A3** of FIG. **6**), and a central modification layer **M3** is formed (process **A4** of FIG. **6**). The peripheral modification layer **M1** serves as a starting point when the peripheral portion **We** is removed in the edge trimming. The internal modification layer **M2** serves as a starting point for separating the processing target wafer **W**. The central modification layer **M3** controls development of cracks in the central portion of the processing target wafer **W**, and serves as a starting point for the separation in the central portion of the processing target wafer **W**.

In the modifying apparatus **60**, the combined wafer **T** is carried into the modifying apparatus **60** by the wafer transfer device **50**, and held on the chuck **100**. Then, the chuck **100** is moved to a formation position of the non-bonding region **Ae**. The formation position of the non-bonding region **Ae** is a position where the laser head **110** is capable of radiating the laser light to the peripheral portion **We** of the processing target wafer **W**.

Then, by radiating laser light **L** (for example, CO₂ laser) from the laser head **110** while rotating the chuck **100** in a circumferential direction thereof, the non-bonding region **Ae** is formed (process **A1** of FIG. **6**). In addition, as described above, the non-bonding region **Ae** can be formed at any position near the bonding interface as long as the bonding

strength between the processing target wafer **W** and the support wafer **S** can be reduced.

Subsequently, the chuck **100** is moved to a macro-alignment position. The macro-alignment position is a position where the macro-camera **120** is capable of imaging an outer end portion of the processing target wafer **W**.

Thereafter, the outer end portion of the processing target wafer **W** is imaged by the macro-camera **120** in 360 degrees in a circumferential direction of the processing target wafer **W**. The obtained image is outputted to the control device **90** from the macro-camera **120**.

In the control device **90**, a first eccentric amount between the center of the chuck **100** and the center of the processing target wafer **W** is calculated from the image obtained by the macro-camera **120**. Further, in the control device **90**, a moving amount of the chuck **100** is calculated based on the first eccentric amount to correct a Y-axis component of the first eccentric amount. The chuck **100** is moved in the Y-axis direction based on the calculated moving amount, and then moved to a micro-alignment position. The micro-alignment position is a position where the micro-camera **121** is capable of imaging the peripheral portion of the processing target wafer **W**. Here, the field of view of the micro-camera **121** is smaller (about $\frac{1}{5}$) than the field of view of the macro-camera **120**, as stated above. Thus, if the Y-axis component of the first eccentric amount is not corrected, the peripheral portion of the processing target wafer **W** may not be included in an angle of view of the micro-camera **121**, resulting in a failure to image the peripheral portion of the processing target wafer **W** with the micro-camera **121**. For the reason, the correction of the Y-axis component based on the first eccentric amount is performed to move the chuck **100** to the micro-alignment position.

Subsequently, a boundary between the bonding region **Ac** and the non-bonding region **Ae** is imaged by the micro-camera **121** in 360 degrees in the circumferential direction of the processing target wafer **W**. The obtained image is outputted to the control device **90** from the micro-camera **121**.

In the control device **90**, a second eccentric amount between the center of the chuck **100** and a center of the bonding region **Ac** is calculated from the image obtained by the micro-camera **121**. Further, in the control device **90**, the position of the chuck **100** with respect to the peripheral modification layer **M1** is decided based on the second eccentric amount such that the center of the bonding region **Ac** and the center of the chuck **100** are coincident with each other.

Then, the chuck **100** is moved to a modification position. The modification position is a position where the laser head **110** radiates laser light to the processing target wafer **W** to form the peripheral modification layer **M1**. Further, in the present exemplary embodiment, the modification position is the same as the micro-alignment position.

Subsequently, as illustrated in FIG. **8** and FIG. **9**, by radiating laser light **L** (for example, YAG laser) from the laser head **110**, the peripheral modification layer **M1** is formed at the boundary between the peripheral portion **We** and the central portion **Wa** of the processing target wafer **W** (process **A2** of FIG. **6**). Further, within the processing target wafer **W**, a crack **C1** develops from the peripheral modification layer **M1** in a thickness direction of the processing target wafer **W**. The crack **C1** reaches the front surface **Wa** but does not reach the rear surface **Wb**.

A lower end of the peripheral modification layer **M1** formed by the laser light **L** is located above a surface of the separated processing target wafer **W** after being finally

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processed. That is, the formation position of the peripheral modification layer M1 is adjusted such that the peripheral modification layer M1 is not left in the first separation wafer W1 after being separated (more specifically, after a grinding processing to be described later).

In the process A2, to locate the chuck 100 at the position decided by the control device 90, the chuck 100 is rotated by the rotating mechanism 103 so that the center of the bonding region Ac and the center of the chuck 100 are coincident, and, also, the chuck 100 is moved in the Y-direction by the moving mechanism 104 (eccentricity correction). At this time, the rotation of chuck 100 and the movement of the chuck 100 in the Y-axis direction are synchronized.

While performing the eccentricity correction of the chuck 100 (processing target wafer W) as described above, the laser light L is radiated to the inside of the processing target wafer W from the laser head 110. That is, while correcting the second eccentric amount, the peripheral modification layer M1 is formed. The peripheral modification layer M1 is formed in a ring shape to be concentric with the bonding region Ac. Accordingly, the peripheral portion We can be appropriately removed later, starting from the peripheral modification layer M1 (crack C1).

Further, in the present exemplary embodiment, if the second eccentric amount includes an X-axis component, this X-axis component is corrected by rotating the chuck 100 while moving it in the Y-axis direction. Meanwhile, if the second eccentric amount does not include the X-axis component, the chuck 100 only needs to be moved in the Y-axis direction without being rotated.

Thereafter, as depicted in FIG. 10 and FIG. 11, by radiating laser light L (for example, YAG laser) from the laser head 110, the internal modification layer M2 is formed along a plane direction of the processing target wafer W (process A3 of FIG. 6). Black arrows shown in FIG. 11 indicate a rotation direction of the chuck 100, and a white arrow indicates a movement direction of a processing point according to the movement of the chuck 100 or the laser head 110. Further, within the processing target wafer W, a crack C2 develops from the internal modification layer M2 along the plane direction. The cracks C2 develop only inside the peripheral modification layer M1 in the diametrical direction.

In addition, if the internal modification layer M2 is formed at a diametrically outer side than the peripheral modification layer M1, the quality of the edge trim after the peripheral portion We is removed may be degraded, as illustrated in FIG. 12. That is, the peripheral portion We may not be appropriately removed starting from the peripheral modification layer M1 (crack C1), and a part of the peripheral portion We may remain on the support wafer S. From this point of view, the formation position of the internal modification layer M2 is adjusted so that it is formed at a diametrically inner side than the peripheral modification layer M1.

Furthermore, a lower end of the internal modification layer M2 formed by the laser light L is located above the surface of the separated processing target wafer W after being finally processed. That is, the formation position of the internal modification layer M2 is adjusted such that the internal modification layer M2 is not left within the first separation wafer after being separated (more specifically after the grinding processing to be described later).

In the process A3, by radiating the laser light L periodically to the inside of the processing target wafer W from the laser head 110 while rotating the chuck 100 (processing target wafer W) and moving the laser head 110 in the Y-axis

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direction from a diametrically outer side of the processing target wafer W toward a diametrically inner side thereof, the internal modification layer M2 is formed in a spiral shape along the plane direction of the processing target wafer W. Details of the method of forming the internal modification layers M2 will be described later.

In the present exemplary embodiment, in forming the internal modification layer M2, the chuck 100 or the laser head 110 is moved in the Y-axis direction. More specifically, between the chuck 100 and the laser head 110, the same member as moved in the eccentricity correction is moved. That is, in the formation of the internal modification layer M2, the eccentricity correction and the formation of the internal modification layer M2 are simultaneously controlled, as will be described later. Further, although the chuck 100 is rotated in forming the internal modification layer M2, the laser head 110 may be moved to rotate the laser head 110 relative to the chuck 100.

In addition, when separating the processing target wafer W starting from the internal modification layer M2 as will be described later, it is desirable that an interval (pitch) of the internal modification layers M2 is set to be uniform in order to perform the separation of the processing target wafer W uniformly within the surface of the processing target wafer W. Thus, in the process A3, by controlling a rotation speed of the chuck 100 and a frequency of the laser light L, the interval between the internal modification layers M2 is adjusted. That is, when a position of the laser head 110 in the diametrical direction (that is, a radiation position of the laser light L) is located at an outer periphery of the processing target wafer W, the rotation speed is reduced, whereas when the position of the laser head 110 in the diametrical direction is located at the central portion of the processing target wafer W, the rotation speed is increased. In addition, when the position of the laser head 110 in the diametrical direction (that is, the radiation position of the laser light L) is located at the outer periphery of the processing target wafer W, the frequency of the laser light L is increased, whereas when the position of the laser head 110 in the diametrical direction is located at the central portion of the processing target wafer W, the frequency is reduced.

If the internal modification layers M2 are formed in the processing target wafer W, the laser light L (for example, YAG laser) is radiated from the laser head 110 to form the central modification layers M3 along the plane direction of the processing target wafer W, as shown in FIG. 13A and FIG. 13B (process A4 of FIG. 6). Within the processing target wafer W, cracks C3 develop from the central modification layers M3 in the plane direction. The central modification layers M3 are formed to be spaced apart from each other (for example, 10 μ m or more) so that the cracks C3 are not connected to each other and are not connected to the cracks C2.

In the process A4, the rotation of the chuck 100 (processing target wafer W) is stopped, and by radiating the laser light L from the laser head 110 to the inside of the processing target wafer W while moving the laser head 110 in horizontal directions (the X-axis direction and the Y-axis direction) above the processing target wafer W, the central modification layer M3 is formed in a straight line shape in the plane direction. Details of this method of forming the central modification layer M3 in the process A4 will be discussed later.

In the formation of the central modification layer M3, the chuck 100 may be moved in the horizontal directions instead of the laser head 110.

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After the central modification layer M3 is formed in the processing target wafer W, the combined wafer T is then carried out of the modifying apparatus 60 by the wafer transfer device 70.

Then, the combined wafer T is transferred into the processing apparatus 80 by the wafer transfer device 70. In the processing apparatus 80, when the combined wafer T is delivered from the transfer ram 71 onto the chuck 83, the processing target wafer W is separated into the first separation wafer W1 and the second separation wafer W2, starting from the peripheral modification layer M1 and the internal modification layer M2 (process A5 of FIG. 6), as illustrated in FIG. 7E. At this time, the peripheral portion We is also removed from the processing target wafer W. Here, since the non-bonding region Ae is formed in the vicinity of the bonding interface between the processing target wafer W and the support wafer S, the peripheral portion We can be easily detached, so that the separation of the processing target wafer W can be carried out appropriately.

In the process A5, the support wafer S is attracted to and held by the chuck 83 while the processing target wafer W is attracted to and held with an attraction surface 71a of the transfer arm 71, as shown in FIG. 14A. Then, as shown in FIG. 14B, the transfer arm 71 is raised in the state that the rear surface Wb of the processing target wafer W is attracted to and held by the attraction surface 71a, so that the processing target wafer W is separated into the first separation wafer W1 and the second separation wafer W2. In the process A5 as stated above, the second separation wafer W2 is separated as one body with the peripheral portion We. That is, the removal of the peripheral portion We and the separation (thinning) of the processing target wafer W are performed at the same time.

The separated second separation wafer W2 is collected to, for example, the outside of the wafer processing system 1. By way of example, a collector (not shown) may be provided within a movable range of the transfer arm 71, and the separated second separation wafer W2 may be collected by releasing the attraction of the second separation wafer W2 in the collector.

In the present exemplary embodiment, the processing target wafer W is separated by using the wafer transfer device 70 in the processing apparatus 80. However, a separation apparatus (not shown) for separating the processing target wafer W may be provided in the wafer processing system 1. This separation apparatus may be stacked on, for example, the modifying apparatus 60.

Next, the chuck 83 is moved to the processing position 80b. Then, as shown in FIG. 7F, a rear surface W1b as a separated surface of the first separation wafer W1 held by the chuck 83 is ground by the grinding unit 84, and the peripheral modification layer M1, the internal modification layer M2, and the central modification layer M3 remaining on the rear surface W1b are removed (process A6 of FIG. 6). In the process A6, by respectively rotating the first separation wafer W1 and the grinding whetstone while keeping the grinding whetstone in contact with the rear surface W1b, the rear surface W1b is ground. Further, the rear surface W1b of the first separation wafer W1 may be then cleaned with a cleaning liquid by using a cleaning nozzle (not shown).

Subsequently, the combined wafer T is transferred to the cleaning apparatus 41 by the wafer transfer device 70. In the cleaning apparatus 41, the rear surface W1b of the first separation wafer W1 as the separated surface is scrub-cleaned (process A7 of FIG. 6). Further, in the cleaning

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apparatus 41, the rear surface Sb of the support wafer S as well as the rear surface W1b of the first separation wafer W1 may be cleaned.

Afterwards, the combined wafer T is transferred to the etching apparatus 40 by the wafer transfer device 50. In the etching apparatus 40, the rear surface W1b of the first separation wafer W1 as the separated surface is wet-etched by a chemical liquid (process A8 of FIG. 6). A grinding mark may be formed on the rear surface W1b ground by the aforementioned processing apparatus 80. In the process A8, the grinding mark can be removed by performing the wet-etching, so that the rear surface W1b can be flattened.

Then, the combined wafer T after being subjected to all the required processings is transferred to the transition device 30 by the wafer transfer device 50, and then transferred into the cassette Ct on the cassette placing table 10 by the wafer transfer device 20. Accordingly, a series of the processes of the wafer processing in the wafer processing system 1 is ended.

In the above-described exemplary embodiment, the processing sequence of the processes A1 to A8 may be appropriately changed.

As a first modification example, the order of the formation of the peripheral modification layer M1 in the process A2 and the formation of the internal modification layer M2 in the process A3 may be reversed. In this case, the wafer processing is performed in the order of the process A1, the process A3, the process A2, and the processes A4 to A8.

As a second modification example, the formation of the central modification layer M3 in the process A4 may be performed before the formation of the peripheral modification layer M1 in the process A2. In this case, the wafer processing is performed in the order of the process A1, the process A4, the processes A2 and A3, and the processes A5 to A8.

As a third modification example, the formation of the central modification layer M3 in the process A4 may be performed before the formation of the internal modification layer M2 in the process A3. In this case, the wafer processing is performed in the order of the processes A1 and A2, the process A4, the process A3, and the processes A5 to A8.

As a fourth modification example, the formation of the non-bonding region Ae in the process A1 may be performed after the formation of the peripheral modification layer M1 in the process A2. In this case, the wafer processing is performed in the order of the process A2, the process A1, and the processes A3 to A8.

As a fifth modification example, the formation of the non-bonding region Ae in the process A1 may be performed after the formation of the internal modification layer M2 in the process A3. In this case, the wafer processing is performed in the order of the processes A2 and A3, the process A1, and the processes A4 to A8.

Further, in the above-described exemplary embodiment, the processings of the processes A1 to A8 may be appropriately omitted.

As a first example of the omission, the removal of the peripheral modification layer M1, the internal modification layer M2 and the central modification layer M3 in the process A6 may be carried out by the wet etching in the process A8. In this case, the grinding processing of the process A6 may be omitted.

As a second example of the omission, if the peripheral modification layer M1, the internal modification layer M2, and the central modification layer M3 are properly removed

and no grinding mark is formed in the grinding processing of the process A6, the wet etching of the process A8 may be omitted.

As a third example of the omission, if the combined wafer T having the non-bonding region Ae formed therein is carried into the wafer processing system 1, the formation of the non-bonding region Ae in the process A1 may be omitted.

Further, if the formation of the non-bonding region Ae is performed after the alignment of the processing target wafer W in the modifying apparatus 60 as in the above-described modification examples 4 and 5, the above-described micro-alignment (calculation of the second eccentric amount between the center of the chuck 100 and the bonding region Ac by imaging the boundary of the non-bonding region Ae) may be omitted. In this case, the formation of the peripheral modification layer M1 in the process A2 may be carried out based on the result of the macro-alignment.

Furthermore, in the process A5 of the above-described exemplary embodiment, the second separation wafer W2 is separated as one body with the peripheral portion We, that is, the removal of the peripheral portion We and the thinning of the processing target wafer W are performed at the same time. However, the second separation wafer W2 and the peripheral portion We do not have to be separated at the same time. For example, the second separation wafer W2 may be separated after the peripheral portion We is removed by the edge trimming. In this case, by allowing the crack C1 developing from the peripheral modification layer M1 formed in the process A2 to reach the front surface Wa and the rear surface Wb as shown in FIG. 15A, the edge trimming processing and the thinning processing can be performed appropriately, as illustrated in FIG. 15B. There may be a case where the peripheral portion We need not be removed. In this case, the alignment of the processing target wafer W may be performed by the outer end of the processing target wafer W instead of the boundary between the bonding region Ac and the non-bonding region Ae.

In addition, in the above-described exemplary embodiment, though both the horizontal (Y-axis direction) movement in forming the internal modification layer M2 and the horizontal (Y-axis direction) movement in performing the eccentricity correction are performed by either the chuck 100 or the laser head 110 on one axis, these horizontal movements may be performed on two axes. By way of example, the laser head 110 may perform the horizontal movement for the formation of the internal modification layer M2, and the chuck 100 may perform the horizontal movement for the eccentricity follow-up. Alternatively, the chuck 100 may perform the horizontal movement for the formation of the internal modification layer M2, and the laser head 110 may perform the horizontal movement for the eccentricity follow-up.

Now, the method of forming the internal modification layer M2 in the process A3 will be described. In the process A3, the internal modification layer M2 is formed in the spiral shape, as stated above. As shown in FIG. 16, an interval between the internal modification layers M2 adjacent in the circumferential direction will be referred to as a circumferential interval P (pulse pitch), and an interval between the internal modification layers M2 adjacent in the diametrical direction will be referred to as a diametrical interval Q (index pitch).

As described above, the internal modification layer M2 needs to be formed at the diametrically inner side than the peripheral modification layer M1 in order to suppress quality deterioration of the edge trim. However, when the centers of

the chuck 100 and the processing target wafer W are not coincident, that is, when the correction of the first and second eccentric amounts by the control device 90 is not properly performed, the modification layers may be formed eccentrically with respect to the processing target wafer W. When the modification layers are formed without taking such eccentricity into consideration, there is a likelihood that the internal modification layer M2 is formed at the diametrically outer side than the peripheral modification layer M1.

Thus, in the modifying apparatus 60, in order to suppress the internal modification layers M2 from being formed at the diametrically outer side than the peripheral modification layer M1, eccentricity correction is performed when forming the modification layers. Such eccentricity correction is carried out by moving the chuck 100 and the laser head 110 in the horizontal directions (X-axis direction and Y-axis direction), for example.

FIG. 17 is an explanatory diagram showing a state of modification layers formed within the processing target wafer W by a first eccentricity correction method.

When the centers of the chuck 100 and the processing target wafer W do not coincide with each other, the non-bonding region Ae is formed eccentrically with respect to the processing target wafer W in the process A1. As stated above, the peripheral modification layer M1 is formed in a ring shape to be concentric with the bonding region Ac (non-bonding region Ae) in the process A2. Accordingly, in the process A3, the internal modification layers M2 are formed in the spiral shape along the peripheral modification layer M1 to be concentric with the bonding region Ac and the peripheral modification layer M1. That is, in the first eccentricity correction method, both the peripheral modification layer M1 and the internal modification layers M2 are formed while the eccentricity correction thereof is performed.

As described above, according to the first eccentricity correction method, by forming the internal modification layers M2 to be concentric with the peripheral modification layer M1 which is formed to follow the eccentricity of the bonding region Ac, the formation of the internal modification layers M2 at the diametrically outer side than the peripheral modification layer M1 can be suppressed.

As described in the first eccentricity correction method, it is desirable that the internal modification layers M2 are formed to follow the eccentricity. If, however, the internal modification layers M2 are formed in the center portion of the processing target wafer W to follow this eccentricity, it is necessary to reciprocate the chuck 100 and the laser head 110 in the horizontal directions at a high speed. As a result, there are concerns that the eccentricity correcting operation may not be able to keep up with the operation of forming the internal modification layers M2, and resonance and guide lifetime may be reduced. Therefore, in a second eccentricity correction method to be described below, the eccentricity correcting operation is not performed at least in the center of the processing target wafer W.

FIG. 18 is an explanatory diagram showing a state of modification layers formed within the processing target wafer W by the second eccentricity correction method.

When the centers of the chuck 100 and the processing target wafer W are not coincident, the non-bonding region Ae is formed eccentrically with respect to the processing target wafer W in the process A1. As stated above, the peripheral modification layer M1 is formed in the ring shape to be concentric with the bonding region Ac (non-bonding region Ae) in the process A2.

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Subsequently, in the second eccentricity correction method, a buffer layer B for absorbing the eccentricity of the bonding region Ac is formed at a diametrically inner side of the processing target wafer W along the peripheral modification layer M1 while correcting the eccentricity of the chuck 100 (processing target wafer W). To elaborate, after the rotation speed of the chuck 100 is rate-controlled (becomes constant) after its rotation is begun, the laser light L is periodically radiated to the inside of the processing target wafer W from the laser head 110 while rotating the chuck 100 (processing target wafer W) one round (360 degrees) at least, as shown in FIG. 19, so that the annular internal modification layer M2 is formed. Then, the laser head 110 is relatively moved inwards in the diametrical direction of the processing target wafer W (Y-axis direction). By forming the internal modification layers M2 in the plane direction in a certain processing width while repeating the formation of the annular internal modification layer M2 and the inward movement of the laser head 110 in the diametrical direction, the internal modification layers M2 as the buffer layer B are formed to be concentric with the bonding region Ac and the peripheral modification layer M1, as illustrated in FIG. 18 and FIG. 19. Moreover, the buffer layer B is formed in the processing width (for example, 500 μm) equal to or larger than the eccentric amount of the bonding region Ac, for example.

Further, the diametrical interval Q of the internal modification layers M2 in the buffer layer B may be set as required.

After the buffer layer B is formed, the internal modification layer M2 is formed in a spiral shape from a certain point within the processing width of the buffer layer B, for example. In the formation of this spiral-shaped internal modification layer M2, the above-described eccentricity correction is not performed. That is, in the second eccentricity correction method, the internal modification layers M2 of the concentric circle shapes forming the buffer layer B and the peripheral modification layer M1 are formed while performing the eccentricity correction, whereas the spiral-shaped internal modification layers M2 formed at the diametrically inner side than the buffer layer B are formed without performing the eccentricity correction.

As described above, according to the second eccentricity correction method, by forming, at the diametrically inside of the peripheral modification layer M1, the buffer layer B which is formed in the processing width equal to or larger than the eccentric amount of the bonding region Ac, it is not necessary to perform the eccentricity correction in the formation of the spiral-shaped internal modification layer M2. That is, even if the internal modification layer M2 is formed eccentrically, since the eccentric amount is absorbed in the processing width of the buffer layer B, the internal modification layer M2 does not extend to the diametrically outer side than the peripheral modification layer M1. Further, since the eccentricity correction need not be performed in the formation of the internal modification layer M2, the internal modification layer M2 can be formed more easily.

Further, by eliminating the need to perform the eccentricity correction in the central portion of the processing target wafer W, the aforementioned failure in properly performing the eccentricity correction can be suppressed. Furthermore, the concerns for the occurrence of resonance and the reduction of the guide lifetime can be reduced. Additionally, since the eccentricity correction is not performed in the central portion as described above, the high rotation speed of the

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chuck 100 can be maintained, and as a result, the circumferential interval P of the internal modification layers M2 can be regulated constant.

FIG. 20 is an explanatory diagram illustrating a state of modification layers formed within the processing target wafer W by a third eccentricity correction method.

When the centers of the chuck 100 and the processing target wafer W are not coincident, the non-bonding region Ae is formed eccentrically with respect to the processing target wafer W in the process A1. The peripheral modification layer M1 is annularly formed in the process A2 to be concentric with the bonding region Ac (non-bonding region Ae) as stated above.

Subsequently, in the third eccentricity correction method, at the diametrically inner side than the peripheral modification layer M1 which is annularly formed to be concentric with the bonding region Ac (non-bonding region Ae) in the process A2, eccentricity correction is performed in a range where the laser head 110 is located at an outer periphery of the processing target wafer W. That is, while moving the laser head 110 from the diametrically outer side toward the diametrically inner side, the chuck 100 is rotated by the rotating mechanism 103 so that the center of the chuck 100 and the center of the bonding region Ac coincide with each other, and, also, the chuck 100 is moved in the Y-axis direction by the moving mechanism 104.

To elaborate, a formation range of the internal modification layer M2 in the processing target wafer W is divided into a plurality of regions along the diametrical direction, and an eccentric stroke is reduced gradually along these regions. FIG. 20 illustrates an example where the formation range of the internal modification layers M2 is divided into a central region R11 as a non-correction region and four annular regions R12 to R15 as an eccentricity correction region, and an eccentric amount of 100 μm is corrected by every 20 μm in each of the annular regions R12 to R15, that is, an example where the eccentric stroke is attenuated by 20 μm .

As described above, according to the third eccentricity correction method, by performing the eccentricity correction in the range included in the outer periphery of the processing target wafer W (the annular regions R12 to R15 in FIG. 20), it is not necessary to perform the eccentricity correction near the central portion of the processing target wafer W. That is, at the outer periphery of the processing target wafer W, the above-described eccentricity correction (attenuation of the eccentric stroke) is completed, so the eccentric amount is 0 μm . In the central portion (central region R11 in FIG. 20), the centers of the chuck 100 and the bonding region Ac coincide with each other. When forming the internal modification layer M2 as described above, the rotation speed of the chuck 100 is low when the laser head 110 is located at the outer periphery of the processing target wafer W. Therefore, the eccentricity correction can be appropriately carried out. As a result, the eccentric amount can be absorbed, and the internal modification layer M2 can be formed inside the peripheral modification layer M1. At this time, since the eccentricity correction is not performed in the central portion, the high rotation speed of the chuck 100 can be maintained, and, as a consequence, the circumferential interval P of the internal modification layer M2 can be controlled constant.

Further, by removing the need to perform the eccentricity correction in the central region R11 of the processing target wafer W, the failure in properly performing the eccentricity correction as described above can be suppressed. Further-

more, the concerns for the occurrence of the resonance and the reduction of the guide lifetime can be reduced.

Moreover, the number of the annular regions for performing the eccentricity correction is not limited to the example of the present exemplary embodiment, and it can be selected as required. In addition, the eccentricity need not necessarily be corrected in a step manner for the annular regions as in the present exemplary embodiment. The eccentricity correction may be continuously performed from the outer periphery of the processing target wafer W toward the center thereof. As an example, the eccentricity correction may be performed for the time during which the laser head 110 radiates the laser light L several rounds from the outer side of the processing target wafer W.

In addition, when correcting the eccentric amount at the outer periphery of the processing target wafer W by the third eccentricity correction method, it is desirable that the eccentricity correction is completed up to the half ($\pi/2$) of a radius of the processing target wafer W. That is to say, it is desirable that the radius of the central region R11 shown in FIG. 20 is equal to or larger than $r/2$.

The internal modification layer M2 in the process A3 is formed as described above. In this way, by performing the eccentricity correction in the formation of the internal modification layer M2, the edge trimming processing and the thinning processing can be easily carried out. Moreover, for this reason, it becomes easy to maintain the quality of the edge trimming processing and the thinning processing, and control in both processings can be simplified.

Now, the method of forming the central modification layer M3 in the process A4 will be elaborated.

As described above, in order to perform the separation of the processing target wafer W uniformly within the surface thereof, it is desirable that a formation interval of the internal modification layers M2 is uniform. To control this formation interval to be constant, the rotation speed of the chuck 100 and the frequency of the laser light L are controlled in the formation of the internal modification layers M2 in the process A3.

However, when the rotation speed of the chuck 100 reaches an upper limit and the frequency of the laser light L reaches a lower limit, the circumferential interval P of the internal modification layers M2 reaches a threshold at which it can be no more enlarged. In this state, if the radiation position of the laser light L is moved inwards in the diametrical direction, the circumferential interval P decreases, so that the internal modification layers M2 may overlap on the same processing line in the central portion of the processing target wafer W. As a result, the central portion of the processing target wafer W may not be properly separated.

The reason why the central portion of the processing target wafer W cannot be separated will be described in further detail. For example, when the internal modification layers M2 are formed to be overlapped, that is, when the radiation of the next (second) laser light L is overlapped with the first formed internal modification layer M2, it becomes difficult for the cracks C2 to develop in the diametrical direction. For another example, even if the internal modification layers M2 are not overlapped, if the circumferential interval P is smaller than a certain threshold, the next (second) laser light L is radiated to the cracks C2 which is developing from the first formed internal modification layer M2. In this case, since the laser light L is radiated to the cracks C2 from which a stress is already released, it becomes difficult again for the cracks C2 to develop in the diametrical direction. For these reasons, since the cracks C2 may not be

properly developed in the central portion of the processing target wafer W, there may arise a case when the central portion of the processing target wafer W may not be separated.

Further, if the internal modification layers M2 are formed to be overlapped as described above, transmitted light of the next (second) laser light L may be generated, and some of the laser light may be transmitted downwards, affecting the device layer D.

Thus, the present inventors have come up with a method of forming the central modification layer M3 as a way to separate the central portion of the processing target wafer W properly. That is, the formation of the internal modification layer M2 is terminated near the central portion of the processing target wafer W where the circumferential interval P reaches the threshold, and the central modification layer M3 is formed at the diametrically inside of the internal modification layer M2, as illustrated in FIG. 21. A formation region R3 of the central modification layer M3 shown in FIG. 21 may be calculated from the minimum value of the frequency of the laser light L and the maximum value of the rotation speed of the chuck 100 (for example, a range of about 1 mm to 2 mm from the center of the processing target wafer W).

In this way, by terminating the formation of the internal modification layer M2 at the required position calculated from the rotation speed of the chuck 100 and the frequency of the laser light, the overlapping of the formed internal modification layers M2 can be suppressed so that the processing target wafer W can be separated properly, and generation of the transmitted light of the laser light L can be suppressed. Further, by forming the central modification layer M3 at the diametrically inside of the internal modification layers M2, the cracks C2 that develop from the internal modification layers M2 can be suppressed from being connected in the central portion of the processing target wafer W to form a protruding portion.

Furthermore, the central modification layer M3 may be formed to have any of various shapes at the diametrically inside of the internal modification layers M2. For example, although the central modification layer M3 is formed by a plurality of, for example, seven straight lines in the example shown in FIG. 21, the shape of the central modification layer M3 may not be limited thereto. By way of example, the central modification layer M3 may be formed by less than 7, for example, as shown in FIG. 22, only one straight line as long as the separation of the processing target wafer W in the central portion thereof can be performed appropriately. As stated above, by reducing the number of the central modification layer M3 to be formed, tact for the formation of the central modification layer M3 can be reduced. Moreover, the shape of the central modification layer M3 is not limited to the straight line shape, and it may be formed to have, for example, only a curved shape or a combination of the straight line shape and the curved shape.

Here, if processing lines of the central modification layers M3 are formed to cross each other, there is a likelihood that the multiple central modification layers M3 may be formed to overlap each other or transmitted light of the laser light may be generated at the corresponding intersection(s). In addition, if the central modification layers M3 are formed to be close to each other, the cracks C3 developing from these central modification layers M3 may be connected to each other, resulting in formation of a protruding portion in the central portion of the processing target wafer W, which in

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turn arises a likelihood that the flatness of the separated surface of the processing target wafer W may be deteriorated.

Thus, it is desirable that the processing lines of the central modification layers M3 are formed independently so as not to cross each other or to be close to each other, as shown in FIG. 21, such that the cracks C3 developing from the adjacent central modification layers M3 along the plane direction are not connected. Desirably, a formation interval of the central modification layers M3 may be equal to or larger than, e.g., 10 μm .

The central modification layers M3 in the process A4 are formed as described above.

According to the third eccentricity correction method, the internal modification layer M2 is formed at the outer periphery of the processing target wafer W to follow the eccentricity of the peripheral modification layer M1. Therefore, formation of the internal modification layer M2 at the diametrically outside of the peripheral modification layer M1 can be appropriately suppressed. Thus, the quality of the edge trimming processing can be maintained.

Meanwhile, since the correction of the eccentric amount is already ended, it is not necessary to perform the eccentricity correction in the central portion of the processing target wafer W. Thus, the rotation speed of the chuck 100 can be maintained, and, accordingly, the circumferential interval P is can be controlled constant. Therefore, the separation of the processing target wafer W can be performed uniformly within the surface thereof.

Furthermore, since the eccentricity correction need not be performed in the central portion of the processing target wafer W as stated above, reciprocating movements of the chuck 100 and the laser head 110 in the central portion of the processing target wafer W are suppressed, so that occurrence of mechanism resonance and reduction of guide lifetime can be suppressed. That is, the lifetime of the apparatus can be prolonged.

In addition, according to the third eccentricity correction method, since the eccentric amount is gradually corrected at the outer periphery of the processing target wafer W, the diametrical interval Q of the internal modification layer M2 within the surface of the processing target wafer W is not sharply changed. That is, the diametrical interval Q of the internal modification layer M2 can be controlled to be substantially constant. Accordingly, the separation of the processing target wafer W can be performed uniformly within the surface of the processing target wafer W more appropriately.

In particular, it has not been conventionally performed to form the internal modification layer M2 while changing the eccentricity correction amount in real time, as in the third eccentricity correction method of the present exemplary embodiment. According to the third eccentricity correction method, throughput for the formation of the modification layers in the processing target wafer W can be improved.

Further, in the present exemplary embodiment, since the internal modification layer M2 is formed in the spiral shape without being cut in the middle within the surface of the processing target wafer W, the tact for the formation of the internal modification layer M2 can be improved.

Further, according to the above-described exemplary embodiment, the internal modification layers M2 are formed such that the circumferential interval P and the diametrical interval Q are uniform within the surface of the processing target wafer W. However, the formation interval of the internal modification layers M2 may not be limited thereto.

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FIG. 23A and FIG. 23B are explanatory diagrams illustrating a method of forming the internal modification layer M2 according to a second exemplary embodiment. As shown in FIG. 23A and FIG. 23B, in the surface of the processing target wafer W in which the internal modification layer M2 according to the second exemplary embodiment is formed, there are formed areas where the diametrical intervals Q between the internal modification layers M2 are different. Specifically, a wide-interval region R1, as a first modification layer formation region, in which the diametrical interval Q between the neighboring internal modification layers M2 is set to be wide is formed at a diametrically outer side of the processing target wafer W, and a narrow-interval region R2, as a second modification layer formation region, in which the diametrical interval Q between the neighboring internal modification layers M2 is set to be narrow is formed at a diametrically inner side than the wide-interval region R1. Further, the circumferential interval P of the internal modification layer M2 is constant over the entire circumference both in the wide-interval region R1 and the narrow-interval region R2.

In addition, in the following description, the internal modification layer M2 formed in the wide-interval region R1 may sometimes be referred to as an outer modification layer M2e, and the internal modification layer M2 formed in the narrow-interval region R2 may sometimes be referred to as an inner modification layer M2c.

Here, in the wide-interval region R1, a formation interval Q1 of the neighboring outer modification layers M2e is set such that the cracks C2 which develop in the plane direction during the formation of these neighboring outer modification layers M2e are not connected to each other, as shown in FIG. 23B. Further, in the narrow-interval region R2, a formation interval Q2 of the neighboring inner modification layers M2c is set so that the cracks C2 which develop in the plane direction during the formation of these neighboring inner modification layers M2c are connected to each other, as shown in FIG. 23B. As an example, the formation interval Q1 of the outer modification layers M2e may be 60 μm , and the formation interval Q2 of the inner modification layers M2c may be 10 μm .

Further, the internal modification layer M2 is formed by radiating the laser light to the inside of the processing target wafer W to amorphize (polycrystallize) the portion to which the laser light is radiated. At this time, in the internal modification layer M2, a compressive stress is generated, as shown in FIG. 24A. Here, in the wide-interval region R1, since the cracks C1 of the adjacent outer modification layers M2e are not connected, the generated compressive stress is accumulated in the outer modification layers M2e. Accordingly, a tensile stress resulting from the compressive stress are accumulated between the outer modification layers M2e adjacent in the diametrical direction, as shown in FIG. 24A. Regions in which the tensile stress acts (hereinafter referred to as the "tensile regions U") are annularly formed over the entire circumference of the processing target wafer W, as shown in FIG. 24B.

Now, a method of forming the wide-interval region R1 and the narrow-interval region R2 as described above, and a method of separating the processing target wafer W will be described. FIG. 25 is a flowchart showing main processes of the method of forming the internal modification layers M2 in the process A3 and the method of separating the processing target wafer W. FIG. 26A to FIG. 26E are explanatory diagrams schematically showing the main processes of the method of forming the internal modification layers M2 in the process A3 and the method of separating the processing

target wafer W. Each of FIG. 26A to FIG. 26E illustrates a cross section of the half of the processing target wafer W in the diametrical direction, seen from a thickness direction thereof. In addition, in FIG. 26A to FIG. 26E, illustration of the support wafer S is omitted for the simplicity of illustration.

In addition, the peripheral modification layer M1 and the cracks C1 are formed in the processing target wafer W prior to the formation of the internal modification layer M2 (process A2 in FIG. 6 and FIG. 25).

As depicted in FIG. 26A, in the formation of the internal modification layer M2, the wide-interval region R1 is first formed (process A3-1 of FIG. 25). By rotating the chuck 100 (processing target wafer W) and moving the laser head 110 in the Y-axis direction from the diametrically outer side of the processing target wafer W toward the diametrically inner side thereof, the wide-interval region R1 is formed sequentially from the diametrically outer side of the processing target wafer W toward the diametrically inner side thereof. The formation interval Q1 of the outer modification layers M2e is, for example, 60 μm . Here, the cracks C2 that develop from the adjacent outer modification layers M2e in the wide-interval region R1 are not connected.

Further, in the present exemplary embodiment as well, when it is necessary to correct the eccentricity of the processing target wafer W in forming the internal modification layer M2, the eccentricity correction (attenuation of the eccentric stroke) is performed at the outer periphery of the processing target wafer W. That is, the eccentricity correction (attenuation of the eccentric stroke) is completed in the formation of the wide-interval region R1, for example.

Here, since the cracks C2 are not connected to each other, the compressive stresses are accumulated in the internal modification layers M2, and the tensile regions U are formed between the adjacent internal modification layers M2, as stated above.

After the wide-interval region R1 is formed, the narrow-interval region R2 is formed, as illustrated in FIG. 26B (process A3-2 of FIG. 25). The narrow-interval region R2 is formed sequentially from the diametrically inner side of the processing target wafer W toward the diametrically outer side thereof by rotating the chuck 100 (processing target wafer W) and moving the laser head 110 in the Y-axis direction from the diametrically inner side of the processing target wafer W toward the diametrically outer side thereof. The formation interval Q2 of the inner modification layers M2c is, for example, 10 μm . Here, the cracks C2 that develop from the adjacent inner modification layers M2c in the narrow-interval region R2 are connected to each other sequentially.

Further, as illustrated in FIG. 26B, in the formation of the narrow-interval region R2 in the process A3-2, the crack C2 of the inner modification layer M2c located on the outermost side of the narrow-interval region R2 and the crack C2 of the outer modification layer M2e located on the innermost side of the wide-interval region R1 are not connected.

In addition, since the eccentricity correction (attenuation of the eccentric stroke) is completed, that is, the center of the chuck 100 and the center of the bonding region Ac are aligned in the formation of the wide-interval region R1 as described above, there is no need to perform the eccentricity correction in the formation of the narrow-interval region R2.

After the narrow-interval region R2 is formed, a starting point modification layer M2s serving as a starting point for the start of the separation of the processing target wafer W is formed, as depicted in FIG. 26C. Specifically, the internal modification layer M2 as the starting point modification

layer M2s is formed between the wide-interval region R1 and the narrow-interval region R2. Accordingly, the crack C2 of the inner modification layer M2c located on the outermost side of the narrow-interval region R2 and the crack C2 of the one outer modification layer M2e positioned on the innermost side of the wide-interval region R1 are connected.

If the starting point modification layer M2s is formed, the cracks C2 of the wide-interval region R1 and the narrow-interval region R2 are connected, so that the compressive stress accumulated in the one outer modification layer M2e of the wide-interval region R1 is released. Then, by this release of the stress, the one outer modification layer M2e swells in a direction in which the first separation wafer W1 and the second separation wafer W2 are detached, as shown in FIG. 26D. That is, at the position where the corresponding one outer modification layer M2e is formed, the first separation wafer W1 and the second separation wafer W2 are detached with the crack C2 as the boundary (process A3-4 of FIG. 25).

In this way, if the first separation wafer W1 and the second separation wafer W2 are detached at the position where the corresponding one outer modification layer M2e is formed, the crack C2 develops outwards in the diametrical direction, as shown in FIG. 26D, while being affected by the force acting in the thickness direction (detachment direction) of the processing target wafer W due to the detachment. As a result, this crack C2 is connected to the crack C2 which is developing from another outer modification layer M2e adjacent thereto (process A3-5 of FIG. 25).

If the cracks C2 of the one outer modification layer M2e and the another outer modification layer M2e are connected, the compressive stress accumulated in the another outer modification layer M2e is released. Then, by this release of the stress, the first separation wafer W1 and the second separation wafer W2 are detached with the crack C2 as the boundary at the position where the another outer modification layer M2e is formed (process A3-5 of FIG. 25).

Then, if the first separation wafer W1 and the second separation wafer W2 are detached in this way at the position where the another outer modification layer M2e is formed, the crack C2 is made to further develop outwards in the diametrical direction while being affected by the force acting in the thickness direction (detachment direction) of the processing target wafer W due to the detachment.

As the development of the crack C2, the release of the compressive stress, and the detachment of the second separation wafer W2 are repeated in a chain reaction in this way, the crack C2 reaches the peripheral modification layer M1, as illustrated in FIG. 26E (process A3-6 of FIG. 25).

If the internal modification layers M2 are formed in the entire surface of the processing target wafer W while the cracks C2 develop as stated above, the formation of the internal modification layers M2 in the process A3 is completed. Then, the central modification layer M3 is formed at the diametrically inside of the inner modification layers M2c (process A4 in FIG. 6 and FIG. 25), and the peripheral portion We and the second separation wafer W2 are removed thereafter (process A5 in FIG. 6 and FIG. 25).

According to the above-described second exemplary embodiment, the internal modification layers M2 are formed in the processing target wafer W, and these internal modification layer M2 are divided into the wide-interval region R1 and the narrow-interval region R2. In the wide-interval region R1, the detachment of the first separation wafer W1 and the second separation wafer W2 takes place in the chain reaction as stated above.

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As stated above, according to the above-described exemplary embodiment, a gap is formed in the thickness direction within the processing target wafer W due to the detachment of the first separation wafer W1 and the second separation wafer W2. That is, since a region in which the first separation wafer W1 and the second separation wafer W2 are not connected is formed within the surface of the processing target wafer W as shown in FIG. 26E, the force required for the subsequent detachment of the second separation wafer W2 is reduced.

In addition, according to the above-described exemplary embodiment, by enlarging the diametrical interval Q of the internal modification layers M2 in the wide-interval region R1, the number of the internal modification layers M2 to be formed can be reduced. Thus, the time required for the formation of the internal modification layers M2 can be reduced, so that the throughput can be further improved.

Further, according to the above-described exemplary embodiment, the first separation wafer W1 and the second separation wafer W2 in the wide-interval region R1 are separated along the crack C2 that naturally develops by the release of the accumulated stress to be used as the starting point of the separation. For this reason, especially in the wide-interval region R1, a smooth separated surface having a periodic structure can be obtained.

FIG. 27A provides a captured image of a separated surface of the processing target wafer W when the diametrical interval Q of the internal modification layers M2 is set to be uniform within the surface of the processing target wafer W, and FIG. 27B shows a captured image of a separated surface of the processing target wafer W when the internal modification layers M2 are formed to have the wide-interval region R1 and the narrow-interval region R2. As can be seen from FIG. 27A and FIG. 27B, by forming the wide-interval region R1 and the narrow-interval region R2 while allowing the cracks C2 to naturally develop by the release of the accumulated stress, the surface roughness after the separation is ameliorated, and the smooth separated surface can be obtained.

Further, in the second exemplary embodiment as well, since the internal modification layer M2 is formed in the wide-interval region R1 to follow the eccentricity, formation of the internal modification layer M2 at the diametrically outside of the peripheral modification layer M1 can be properly suppressed. Further, since the eccentricity correction (attenuation of the eccentric stroke) is completed in the formation of the wide-interval region R1 and there is no need to perform the eccentricity correction in the narrow-interval region R2, the circumferential interval P can be regulated constant even in the central portion of the processing target wafer W. As a consequence, the separation of the processing target wafer W can be performed uniformly within the surface thereof.

Moreover, according to the above-described second exemplary embodiment, the wide-interval region R1 is formed at the diametrically outer side, and the narrow-interval region R2 is formed at the diametrically inner side. As shown in FIG. 28A, however, the narrow-interval region R2 may be formed at the diametrically outer side of the processing target wafer W, and the wide-interval region R1 may be formed inside this narrow-interval region R2, when viewed from the top. As another example, as shown in FIG. 28B, the wide-interval region R1 and the narrow-interval region R2 may be alternately formed at the diametrically outer side of the processing target wafer W.

Further, in the above-described second exemplary embodiment, the wide-interval region R1 and the narrow-

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interval region R2 are formed with respect to the diametrical direction of the processing target wafer W, that is, the diametrical interval Q of the internal modification layers M2 is changed. Instead, however, the circumferential interval P (pulse pitch) may be changed. Moreover, both the diametrical interval Q and the circumferential interval P may be changed. In such a case, since the number of the internal modification layers M2 to be formed within the surface of the processing target wafer W is further reduced, the throughput can be further improved.

Further, in the above-described second exemplary embodiment, the separation of the processing target wafer W is started by forming the starting point modification layer M2s between the wide-interval region R1 and the narrow-interval region R2. However, the way how to start the separation of the processing target wafer W is not limited thereto. By way of example, after forming the wide-interval region R1 up to a predetermined position from the diametrically outer side of the processing target wafer W toward the diametrically inner side thereof as shown in FIG. 29A, the separation may be begun by forming the narrow-interval region R2 from the center of the processing target wafer W toward the diametrically outer side thereof to be joined with the internal modification layer M2 of the wide-interval region R1, as illustrated in FIG. 29B.

It should be noted that the above-described exemplary embodiment is illustrative in all aspects and is not anyway limiting. The above-described exemplary embodiment may be omitted, replaced and modified in various ways without departing from the scope and the spirit of claims.

According to the exemplary embodiment, it is possible to perform a periphery removing processing and a separation processing for a processing target object appropriately.

We claim:

1. A processing apparatus configured to process a processing target object, comprising:

a holder configured to hold the processing target object;
a modifying device configured to radiate laser light to an inside of the processing target object to form multiple internal modification layers along a plane direction of the processing target object; and

a controller configured to control an operation of forming the internal modification layers in the processing target object,

wherein the controller controls the operation of forming the internal modification layers such that the internal modification layers are formed in a spiral shape from a diametrically outer side of the processing target object toward a diametrically inner side thereof, and such that an eccentric amount between the holder and the processing target object held by the holder in the forming of the internal modification layers toward the diametrically inner side of the processing target object becomes smaller than the eccentric amount in forming the internal modification layers toward the diametrically outer side of the processing target object.

2. The processing apparatus of claim 1,

wherein the processing target object comprises:
an eccentricity correction region in which the eccentric amount is corrected in the forming of the internal modification layers; and

a non-correction region which is formed at a diametrically inside of the eccentricity correction region and in which correction of the eccentric amount is not performed in the forming of the internal modification layers.

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3. The processing apparatus of claim 2,
wherein the non-correction region is formed in a central
portion of the processing target object, and
the controller controls an operation of correcting the
eccentric amount in the non-correction region such that
a center of the holder and a center of the processing
target object held by the holder are coincident. 5
4. The processing apparatus of claim 3,
wherein the non-correction region is formed in a shape of
a circle, and
a radius of the circle forming the non-correction region is
equal to or larger than at least $\frac{1}{2}$ of a radius of the
processing target object. 10
5. The processing apparatus of claim 2,
wherein the eccentricity correction region is divided into
multiple annular regions, and
the controller forms the internal modification layers
sequentially from the diametrically outer side of the
processing target object toward the diametrically inner
side thereof, and controls an operation of correcting the
eccentric amount such that the eccentric amount is
gradually reduced in each of the multiple annular
regions. 15
6. A processing method of processing a processing target
object, comprising:
forming, by a modifying device, internal modification 25
layers in a spiral shape by radiating laser light to an
inside of the processing target object held by a holder;
and
forming the internal modification layers from a diametri-
cally outer side of the processing target object toward 30
a diametrically inner side thereof such that an eccentric
amount between the holder and the processing target
object held by the holder in the forming of the internal
modification layers toward the diametrically inner side
of the processing target object becomes smaller than 35
the eccentric amount in forming the internal modifica-
tion layers toward the diametrically outer side of the
processing target object.

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7. The processing method of claim 6,
wherein the processing target object comprises:
an eccentricity correction region in which the eccentric
amount is corrected in the forming of the internal
modification layers; and
a non-correction region which is formed at a diametrically
inside of the eccentricity correction region and in which
correction of the eccentric amount is not performed in
the forming of the internal modification layers.
8. The processing method of claim 7,
wherein a center of the holder and a center of the
processing target object held by the holder are coinci-
dent in the non-correction region formed in a central
portion of the processing target object.
9. The processing method of claim 8,
wherein the non-correction region is formed in a shape of
a circle having a radius equal to or larger than at least
 $\frac{1}{2}$ of a radius of the processing target object.
10. The processing method of claim 7,
wherein the eccentricity correction region is divided into
multiple annular regions,
the internal modification layers are formed sequentially
from the diametrically outer side of the processing
target object toward the diametrically inner side
thereof, and
the eccentric amount is gradually reduced in each of the
multiple annular regions.
11. The processing method of claim 6,
wherein in the forming of the internal modification layers,
the spiral-shaped internal modification layers are
formed in a plane direction by radiating the laser light
to the inside of the processing target object from the
modifying device periodically while relatively rotating
the processing target object held by the holder and by
moving the modifying device in a diametrical direction
relative to the holder.

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